



CLOUD COMPUTING

UNIT 1

SYLLABUS

Cloud Computing	
Course Code: BIT 304	Credits: 4
Contact Hours: L-3 T-0 P-2	Semester: 6
Course Category: DCC	

Course Objectives:

- To learn the use of various cloud computing services and cloud deployment models.
- Understand the concept of virtualization in cloud computing.
- To apply the concepts of cloud computing for designing, evaluating, simulating and comparing various applications in a cloud computing environment.
- To gain the confidence in resource management and load balancing algorithms in a cloud computing environment.
- To gain the confidence of security attacks and their provisions at various levels of cloud computing.

UNIT I	10 hours
Cloud Computing Fundamentals: Introduction of cloud computing, History of cloud computing, Trends in computing, Grid computing, Cluster computing, Distributed computing, Utility computing, Fog computing, NIST definition and characteristics of cloud computing, Cloud as green and smart, Cloud as IaaS, PaaS, SaaS, BPaaS and HaaS, SPI framework, SPI vs. traditional IT Model, Cloud deployment models, Benefits and challenges.	

Cloud Computing Introduction

Cloud refers to a Network or Internet.

Cloud is something, which is present at remote location. It can provide services over network, that is, on public networks or on private networks, that is, Wide Area Networks (WANs), Local Area Networks (LANs), or Virtual Private Networks (VPNs).

Applications such as e-mail, web conferencing, customer relationship management (CRM), all run in cloud.

Cloud computing is the delivery of computing services over the internet, or "the cloud". It allows users to access computing resources on demand, such as storage, servers, and software, without the need to manage physical infrastructure.



Figure 1: Examples of Cloud Computing

Cloud Computing Introduction

- Cloud is a construct (infrastructure) that allows to access application that actually resides at a remote location of another internet connected device, most often, this will be a distant datacenter.
- Cloud computing takes the technology, services, and applications that are similar to those on the Internet and turns them into a self-service utility.
- Cloud provides an abstraction based on the notion of pooling physical resources and presenting them as a virtual resource.
- It reduces the cost of using that software along with the reduction in the cost of installation of heavy servers. Additionally, cloud aids in reducing the cost of electricity bills.

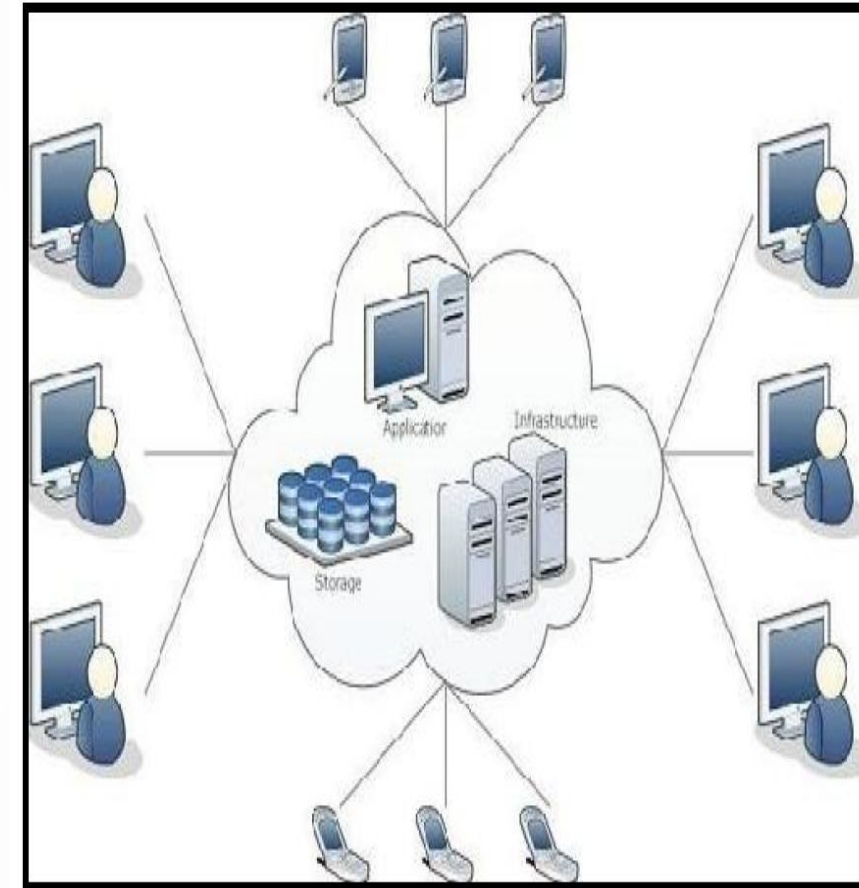


Figure 2: Cloud Scenario

Cloud Properties

- ▶ **Cloud computing is user-centric:** Once you as a user are connected to the cloud, whatever is stored there—documents, messages, images, applications, whatever—becomes yours. In addition, not only is the data yours, but you can also share it with others.
- ▶ **Cloud computing is task-centric:** Instead of focusing on the application and what it can do, the focus is on what you need done and how the application can do it for you.
- ▶ **Cloud computing is powerful:** Connecting hundreds or thousands of computers together in a cloud creates a wealth of computing power impossible with a single desktop PC. You're not limited to a single source of data, as you are with a desktop PC.

Cloud Properties

- **Cloud computing is intelligent:** With all the various data stored on the computers in a cloud, data mining and analysis are necessary to access that information in an intelligent manner.
- **Cloud computing is programmable:** Many of the tasks necessary with cloud computing must be automated. For example, to protect the integrity of the data, information stored on a single computer in the cloud must be replicated on other computers in the cloud. If that one computer goes offline, the cloud's programming automatically redistributes that computer's data to a new computer in the cloud.

History of Cloud Computing

Client Server Architecture: Before emerging the cloud computing, there was Client/Server computing which is basically a centralized storage in which all the software applications, all the data and all the controls are resided on the server side. If a single user wants to access specific data or run a program, he need to connect to the server and then gain appropriate access, and then he can do his business.

Distributed computing: Where all the computers are networked together and share their resources when needed. On the basis of above computing, there was emergence of the concept of cloud computing.

Cloud computing was invented in the early 1960s by J.C.R Licklider, an American Psychologist and Computer Scientist. During his network research work on ARPANet (Advanced Research Project Agency Network), trying to connect people and data all around the world, he gave an introduction to cloud computing technique which we all know today.

History of Cloud Computing

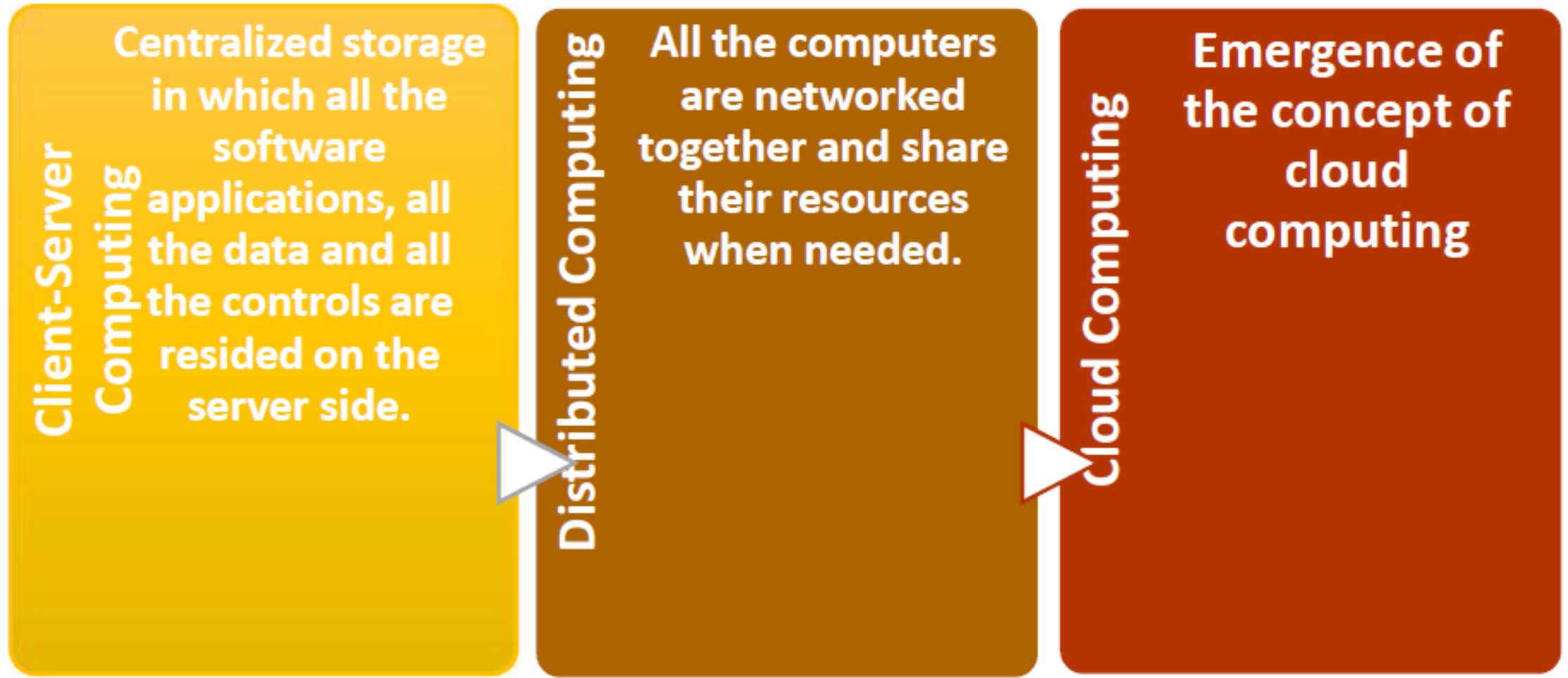


Figure 4: Convergence to Cloud Computing

History of Cloud Computing

- At around in **1961, John MacCharly** suggested in a speech at MIT that **computing can be sold like a utility**, just like a water or electricity.
- In 1999, **Salesforce.com started delivering of applications to users using a simple website**. The applications were delivered to enterprises over the Internet, and this way the dream of computing sold as utility were true.
- The beauty of the cloud computing phase went on running throughout the era of the 21st Century.
- In 2002, **Amazon started Amazon Web Services, providing services like storage, computation and even human intelligence**. However, only starting with the launch of the Elastic Compute Cloud in 2006 a truly commercial service open to everybody existed.
- By 2008, Google too introduced its beta version of the search engine.

History of Cloud Computing

- In 2009, **Google Apps also started to provide cloud computing enterprise applications.**
- Earlier announced by Microsoft in the year 2008, it released its cloud computing service named Microsoft Azure for testing, deployment and managing applications and services.
- In the year 2012, Google Compute Engine was released but was rolled to the public.
- By the end of Dec 2013, Oracle introduced Oracle Cloud with three primary services for business (IaaS, PaaS and SaaS). Currently, as per records, Linux and Microsoft Azure share most of their work parallel.

Evolution of Cloud Computing

- Growth of cloud computing is not an instantaneous task but has transited from several intermediate stages.
- Beginning from an era of mainframe computing where huge and powerful mainframe systems supported many users connected through dummy terminals and running data management applications.
- There exist five intermediary stages from mainframe computing to the use of personal stand-alone computers running desktop applications deriving personal computing to the influx of interconnected computers converging to networking computing. This stage saw the growth of local networks connected to other networks creating a globally interconnected network such as the Internet for utilizing remote applications and resources.
- The networked computers usually functioned in autonomic fashion resulting in autonomic computing or followed client-server architectures resulting in client-server computing.

Evolution of Cloud Computing

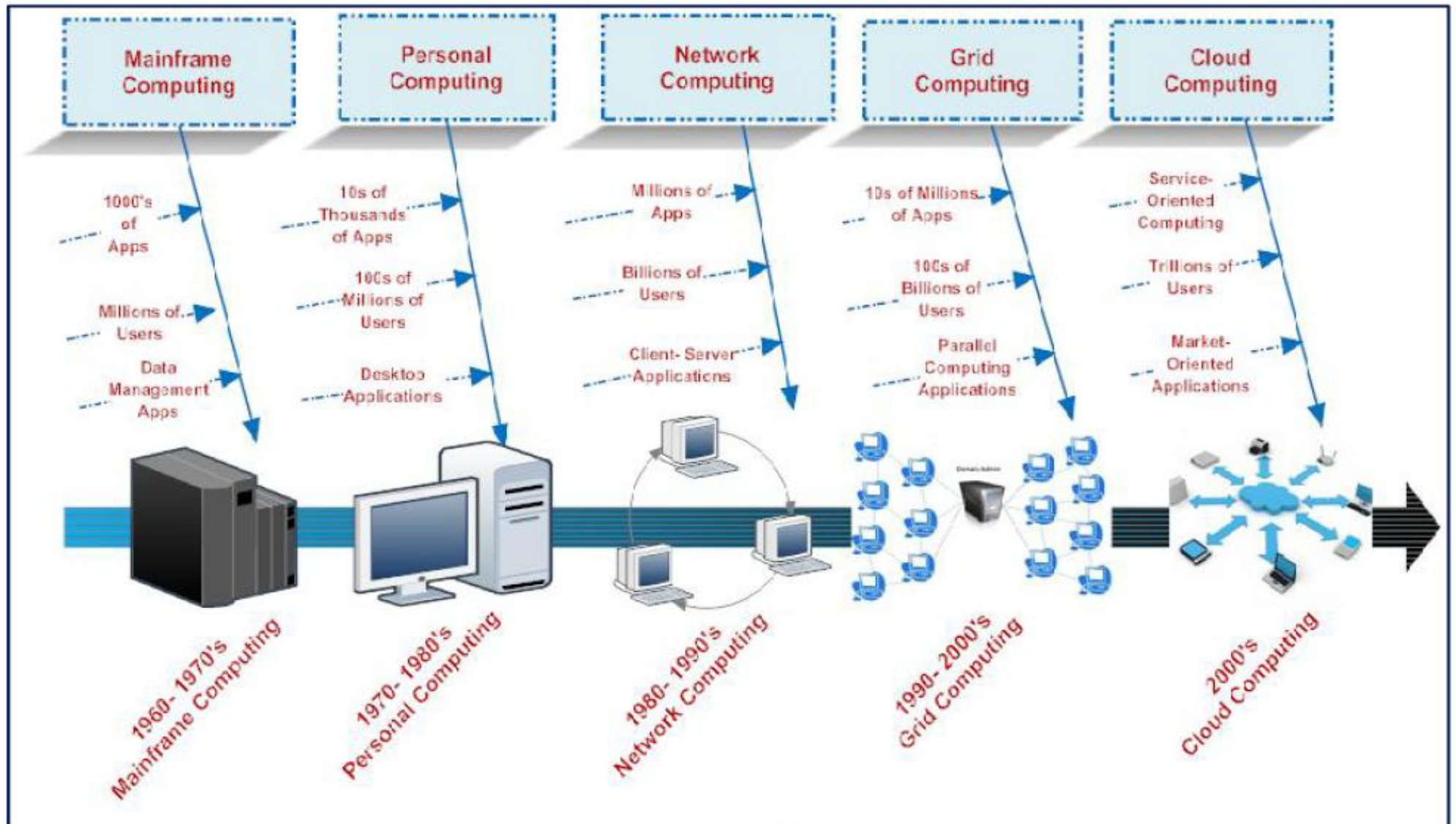


Figure 5: Evolution of Cloud Computing

Evolution of Cloud Computing

Mainframe Computing: A mainframe is a powerful computer that connects to multiple end clients so that users can access apps concurrently.

Mainframe computing refers to the use of large, powerful computers to handle and process vast amounts of data. These computers are designed to support high volumes of transactions, offer large-scale storage, and maintain high reliability, availability, and security.

Despite the rise of distributed computing and cloud services, mainframes continue to be essential for legacy systems and high-performance computing needs.

1. Banking and Financial Services: Transaction Processing, Core Banking Systems.
2. Government and Public Sector: Tax Processing, Census Data Processing
3. Healthcare: Medical Records Management, Insurance Claims Processing
4. Transportation and Logistics: Airline Reservation Systems

Evolution of Cloud Computing

Personal Computing:

- Personal computing refers to the use of computers by individuals for everyday tasks, personal productivity, entertainment, and communication.
- It typically involves desktop PCs, laptops, tablets, and smartphones, which are designed for personal use rather than large-scale, enterprise-level operations.
- Personal computers are equipped with user-friendly interfaces and can run a variety of software applications for tasks such as word processing, web browsing, gaming, graphic design, and social media.
- With the evolution of personal computing, these devices have become more powerful, compact, and interconnected, often featuring cloud computing capabilities, which enable users to store and access data remotely.
- Personal computing has revolutionized how people work, learn, and interact with the world, providing access to vast information and resources at one's fingertips.

Evolution of Cloud Computing

Network Computing:

- Network computing refers to the use of a network of computers and devices to share resources, data, and applications across multiple locations.
- It enables users to connect, communicate, and collaborate by leveraging shared services and centralized systems.
- This approach relies on technologies like local area networks (LANs), wide area networks (WANs), and the internet to connect devices.
- Key features of network computing include **data sharing**, **remote access**, **distributed processing**, and **cloud computing**. Users can access applications and data stored on remote servers rather than relying solely on local devices.
- This model improves efficiency, scalability, and flexibility while reducing the need for extensive hardware on individual devices. Examples of network computing include web browsing, email, file sharing, and enterprise systems like customer relationship management (CRM) and enterprise resource planning (ERP).

Evolution of Cloud Computing

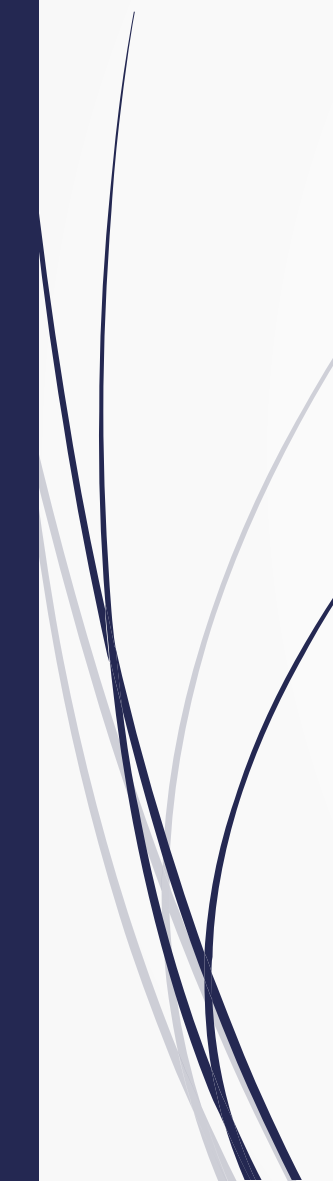
Grid Computing:

- Grid computing is a distributed architecture that combines computer resources from different locations to achieve a common goal. It breaks down tasks into smaller subtasks, allowing concurrent processing.
- It enables sharing of computational resources (such as processing power, storage, and data) across networks to perform tasks that would be too large or resource-intensive for a single computer.
- Grid Computing is a subset of distributed computing, where a virtual supercomputer comprises machines on a network connected by some bus, mostly Ethernet or sometimes the Internet.
- It can also be seen as a form of Parallel Computing where instead of many CPU cores on a single machine, it contains multiple cores spread across various locations. The concept of grid computing isn't new, but it is not yet perfected as there are no standard rules and protocols established and accepted by people.



Evolution of Cloud Computing

Need for Grid Computing:

1. **Handling Large-Scale Problems:** Many modern problems (e.g., climate simulations) require significant computational power that cannot be provided by a single machine.
 2. **Cost-Effective Resource Sharing:** Grid computing enables organizations to share computing resources without needing to invest in large-scale infrastructure.
 3. **Improved Efficiency:** It enables parallel processing, making tasks that once took days or weeks significantly faster.
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Evolution of Cloud Computing

Working of Grid Computing

A Grid computing network mainly consists of these three types of machines

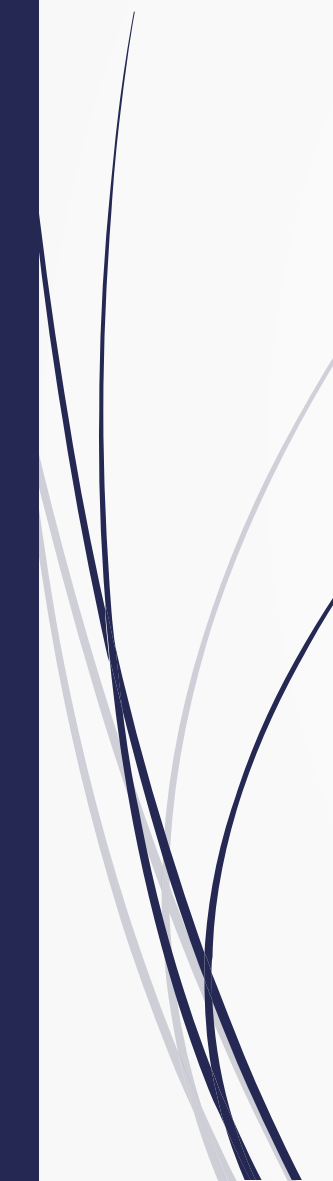
1. **Control Node:** A computer, usually a server or a group of servers which administrates the whole network and keeps the account of the resources in the network pool.
2. **Provider:** The computer contributes its resources to the network resource pool.
3. **User:** The computer that uses the resources on the network.

When a computer makes a request for resources to the control node, the control node gives the user access to the resources available on the network. When it is not in use it should ideally contribute its resources to the network. Hence a normal computer on the node can swing in between being a user or a provider based on its needs.



Evolution of Cloud Computing

Use Cases of Grid Computing

1. Genomic Research
 2. Drug Discovery
 3. Cancer Research
 4. Weather Forecasting
 5. Risk Analysis
 6. Computer-Aided Design (CAD)
 7. Animation and Visual Effects
 8. Collaborative Projects
- 



Evolution of Cloud Computing

Advantages of Grid Computing

1. Grid Computing provide high resources utilization.
2. Grid Computing allow parallel processing of task.
3. Grid Computing is designed to be scalable.

Disadvantages of Grid Computing

1. The software of the grid is still in the evolution stage.
2. Grid computing introduce Complexity.
3. Limited Flexibility
4. Security Risks

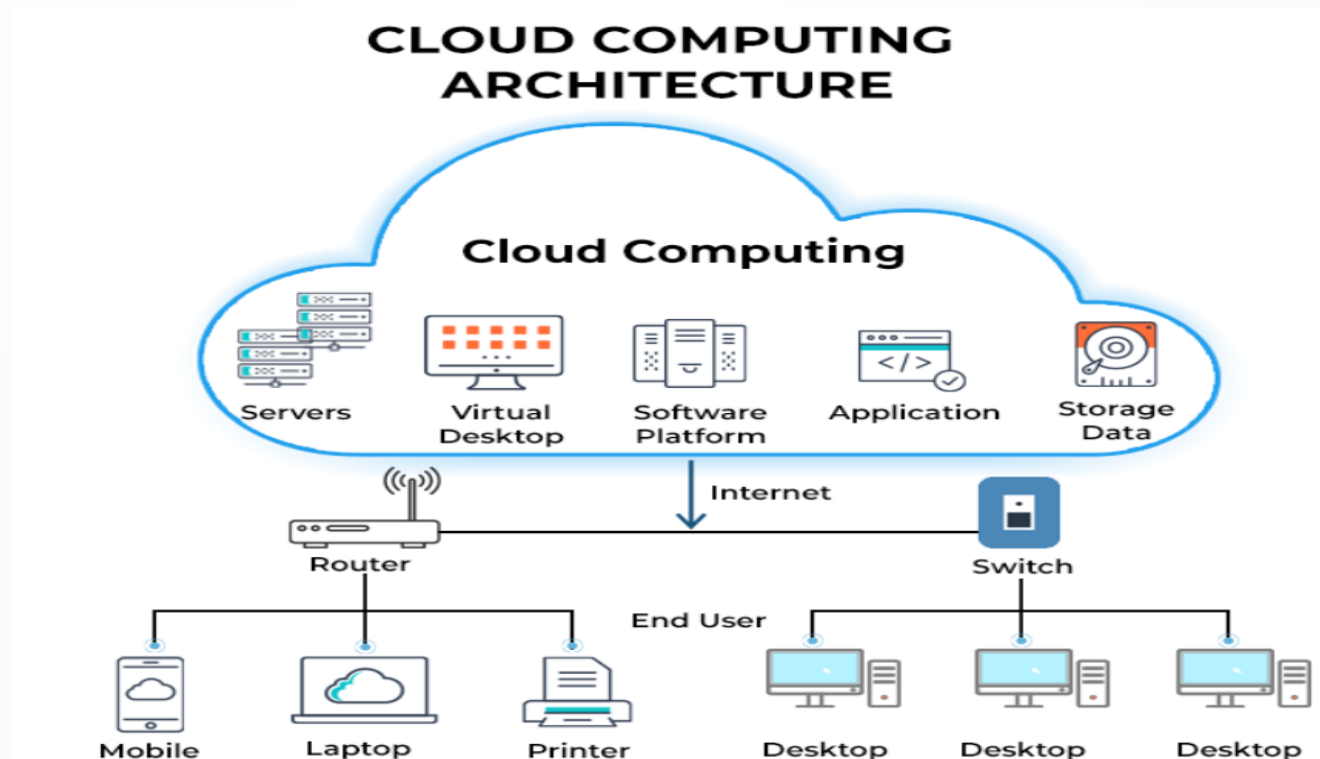
Evolution of Cloud Computing

Aspect	Grid Computing	Distributed Computing
Scope	Large-scale, often cross-organizational tasks	Local or within an organization/network
Resource Sharing	Resources are often heterogeneous and shared across organizations	Resources are typically homogeneous and centralized
Connectivity	Operates over WAN, including the internet	Operates primarily over LAN or private networks
Purpose	High-performance computing (scientific, data-heavy tasks)	General-purpose tasks (e.g., file storage, web services)
Task Distribution	Tasks are divided for parallel processing across many nodes	Tasks are distributed across systems to improve efficiency
Fault Tolerance	Resilient to failures, but can be affected by distributed nature	Designed for fault tolerance but more controlled environment

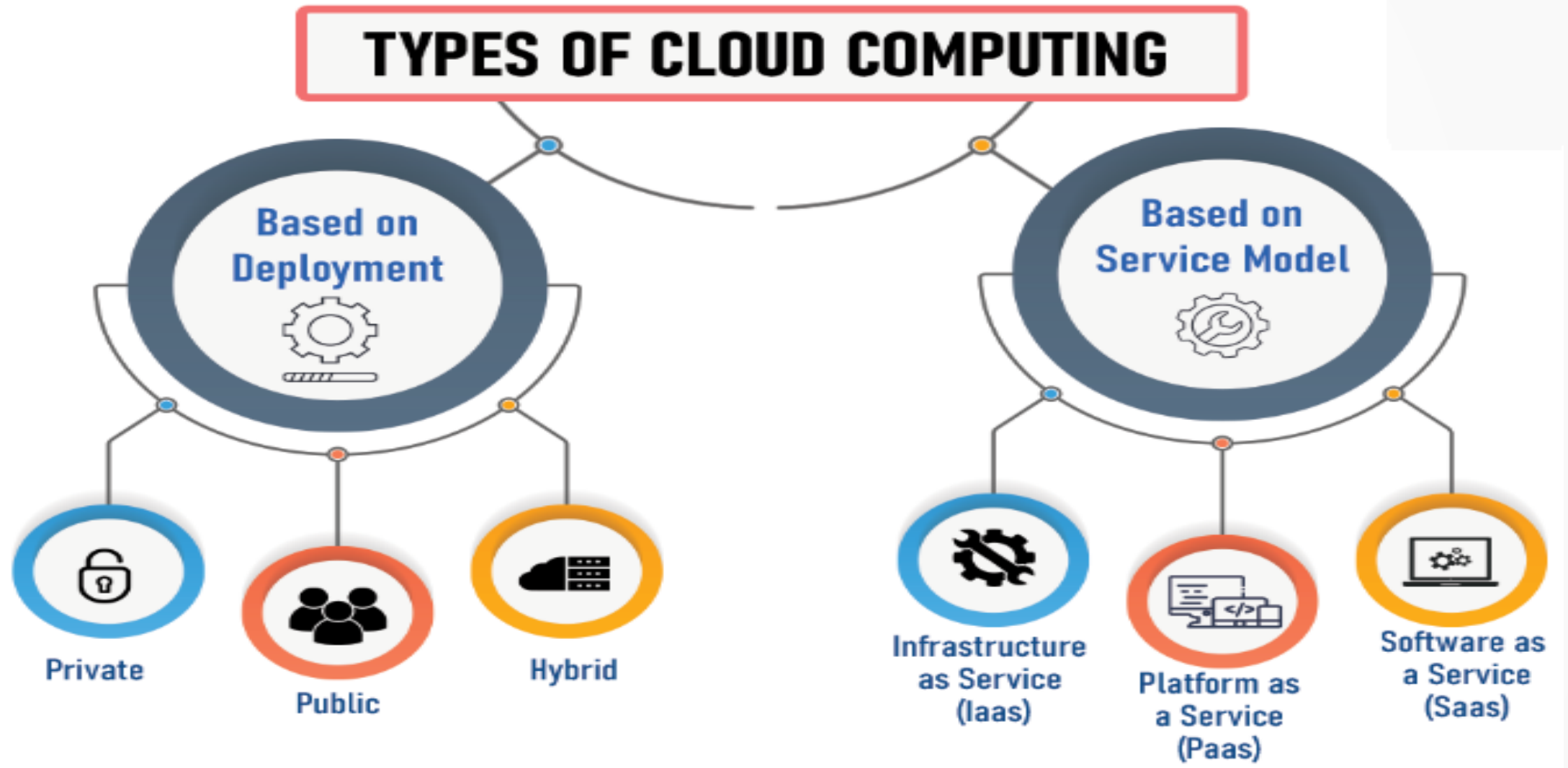
Evolution of Cloud Computing

Cloud computing is defined as the use of hosted services, such as data storage, servers, databases, networking, and software over the internet.

Since cloud computing began, the world has witnessed an explosion of cloud-based applications and services in IT, which continue to expand. Almost every application we use resides on the cloud, helping us save storage space, expenses, and time.



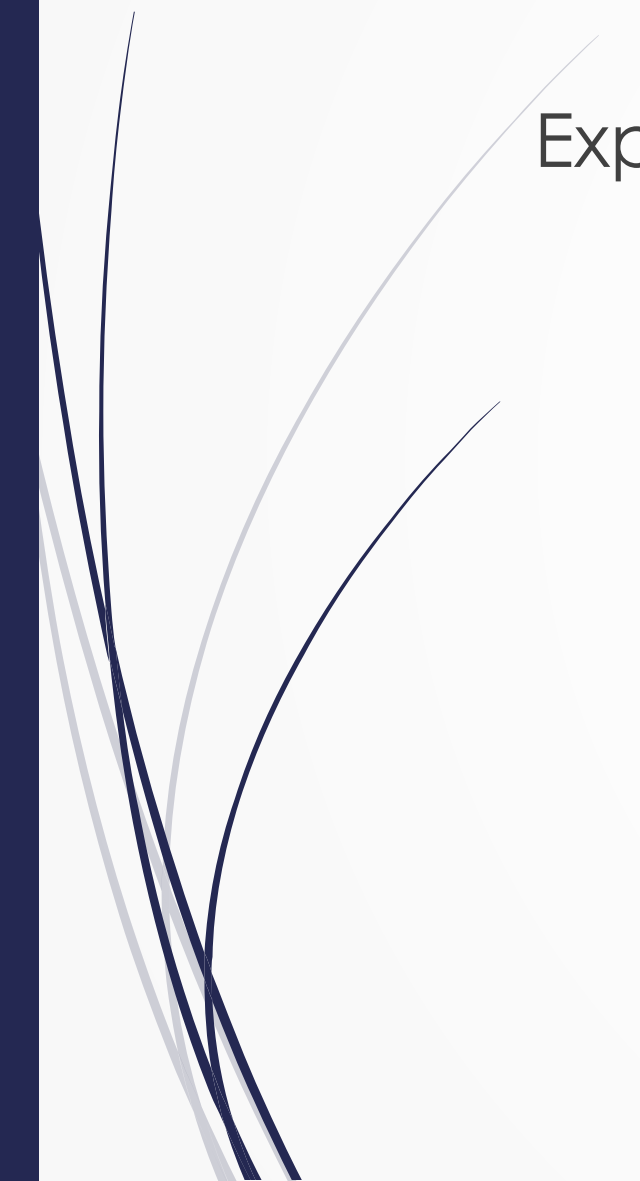
Evolution of Cloud Computing





Class Assignment:

Explore about the Key Differences Between Grid Computing and Cloud Computing

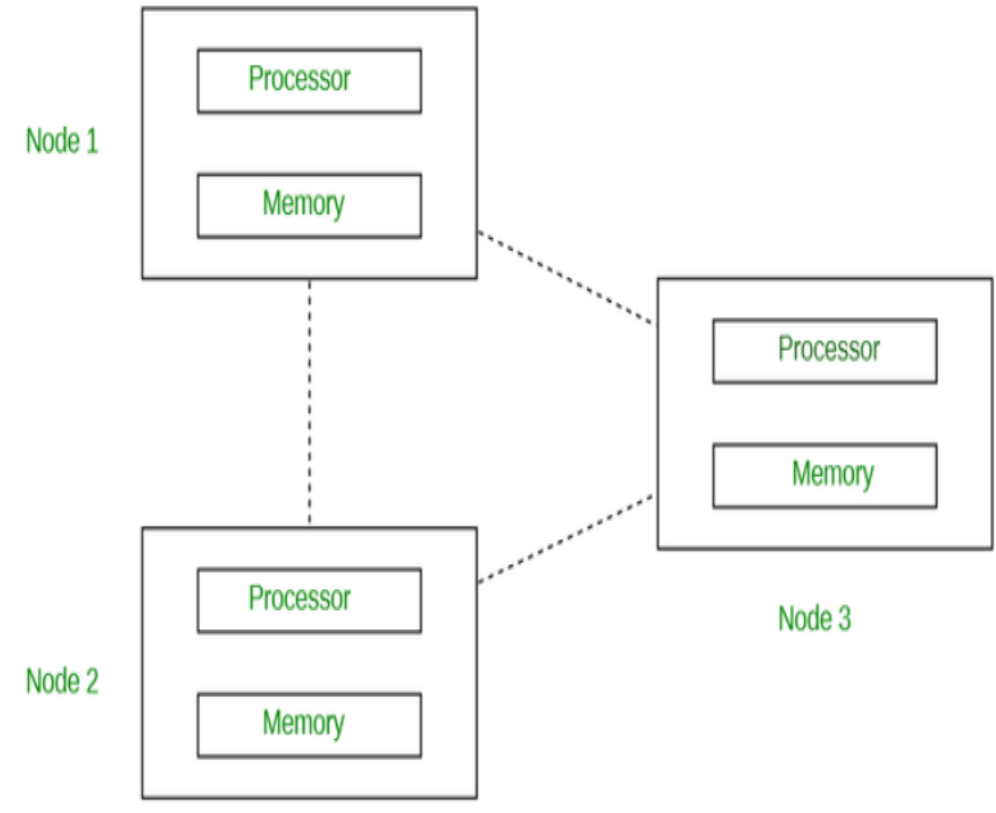
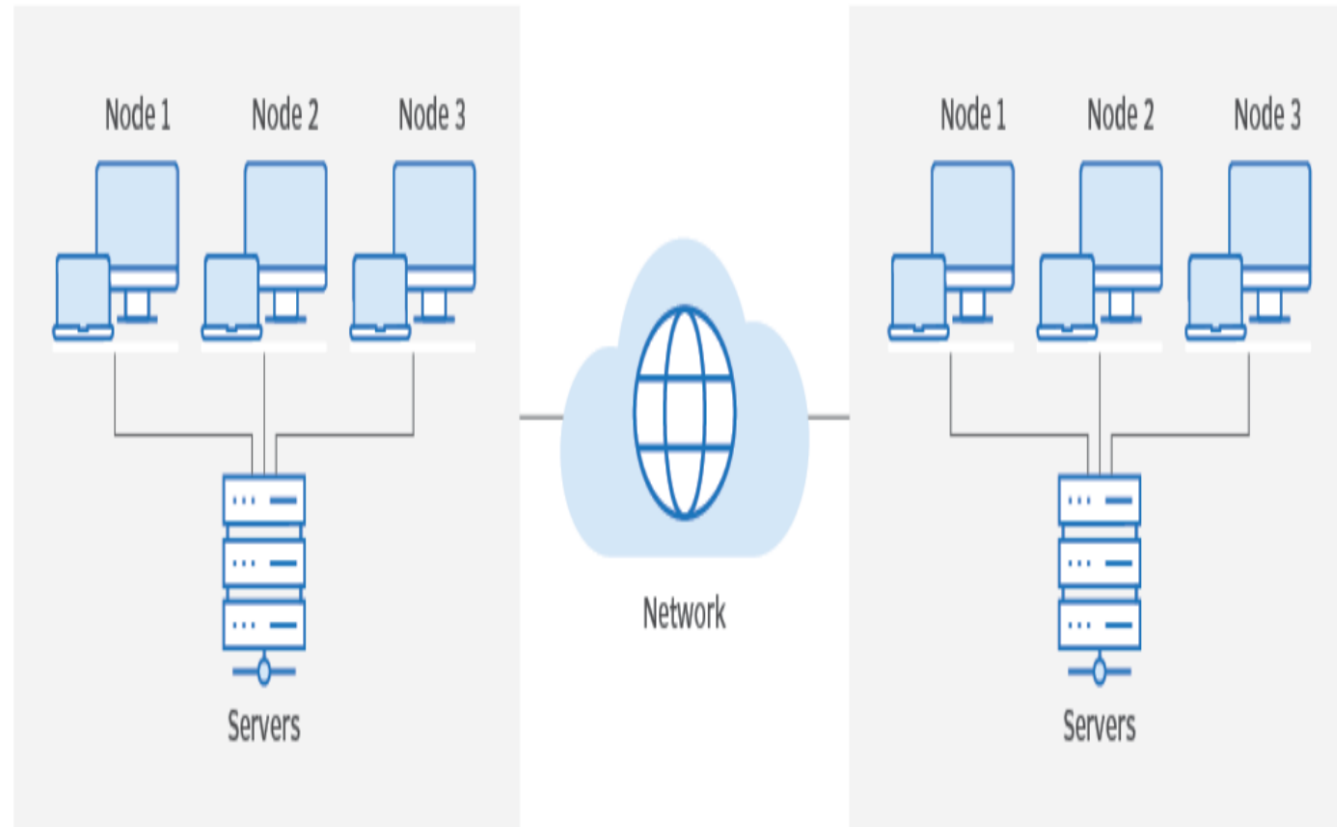


Different Computing Paradigms

Distributed Computing

- A distributed computing is a network of **independent computers working together** to achieve common computational goals.
- It is a system where multiple computers collaborate to solve a problem that is beyond their individual computing capabilities.
- **Each system, or 'node', is self-sufficient**, meaning it operates independently while also contributing to the overall goal.
- In a distributed computing system, **the nodes communicate with each other through various forms of messaging like sending data, signals, or instructions**.
- This communication allows the network of computers to operate as a coherent system, despite each node working independently.

Distributed Computing



Advantages of Distributed Computing?

Scalability :

- As the computational needs of a task increase, instead of upgrading a single system to handle the increased workload, additional nodes can be added to the distributed network.
- This way, the system can efficiently handle the growing demands without major modifications or significant costs.
- Furthermore, scalability in a distributed computing system is not just limited to adding more nodes.
- It also includes the ability to enhance the computational power of existing nodes or to replace older nodes with more powerful ones. This flexibility makes distributed computing an ideal solution for tasks that have unpredictable or rapidly changing computational requirements.

Advantages of Distributed Computing?

Availability

- High availability is another significant advantage of distributed computing. Since the system is composed of multiple independent nodes, the failure of one or a few nodes does not halt the entire system. Other nodes in the network can continue their operations, ensuring that the system as a whole remains functional.
- Moreover, in many distributed computing systems, redundancy is built into the architecture. This means that if one node fails, its tasks can be reassigned to another node, ensuring that there is no loss of service.
- This high availability makes distributed computing systems extremely resilient and reliable.

Advantages of Distributed Computing?

Efficiency

- Distributed computing systems are highly efficient. By dividing a large task into smaller subtasks and processing them concurrently, the system can significantly reduce the time required to complete the task.
- This parallel processing capability is especially beneficial for complex computational tasks that would take an unfeasibly long time to complete on a single computer.
- Moreover, distributed computing systems can optimize resource utilization. Instead of leaving a powerful computer idle while a less powerful one struggles with a task, the system can distribute the workload according to the capabilities of each node.
- This ensures that all resources are used optimally, further enhancing the efficiency of the system.

Advantages of Distributed Computing?

Transparency

- Transparency is a key feature of distributed computing systems. Despite being composed of multiple independent nodes, the system operates as a single entity from the user's perspective.
- This means that the complexities of the underlying architecture, such as the division of tasks, the communication between nodes, and the handling of failures, are hidden from the user.
- This transparency simplifies the user interface and makes the system easier to use.
- It also means that changes to the system, such as the addition or removal of nodes, can be made without affecting the user experience.

Types of Distributed Computing Architecture

Client-Server Architecture: It is a common type of distributed computing architecture.

In this model, the system is divided into two types of nodes: clients and servers. Clients request services, and servers provide them. The servers are typically powerful computers that host and manage resources, while the clients are usually less powerful machines that access these resources.

Three-Tier Architecture: Three-tier architecture is a type of client-server architecture where the system is divided into three layers: the presentation layer, the application layer, and the data layer.

The presentation layer handles the user interface, the application layer processes the business logic, and the data layer manages the database. By separating these functions, the system can achieve greater scalability, flexibility, and maintainability.

Types of Distributed Computing Architecture

N-Tier Architecture : It is a further extension of the three-tier architecture. In this model, the system is divided into 'n' tiers or layers, where 'n' can be any number greater than three.

Each layer is dedicated to a specific function, such as user interface, business logic, data processing, data storage, etc. This division of labor allows for greater modularity, making the system more scalable and easier to manage.

Peer-to-Peer Architecture: Peer-to-Peer (P2P) architecture is a type of distributed computing architecture where all nodes are equal, and each node can function as both a client and a server.

In this model, there is no central server; instead, each node can request services from and provide services to other nodes. This decentralization makes P2P architectures highly scalable and resilient, as there is no single point of failure.

Types of Distributed Computing Architecture

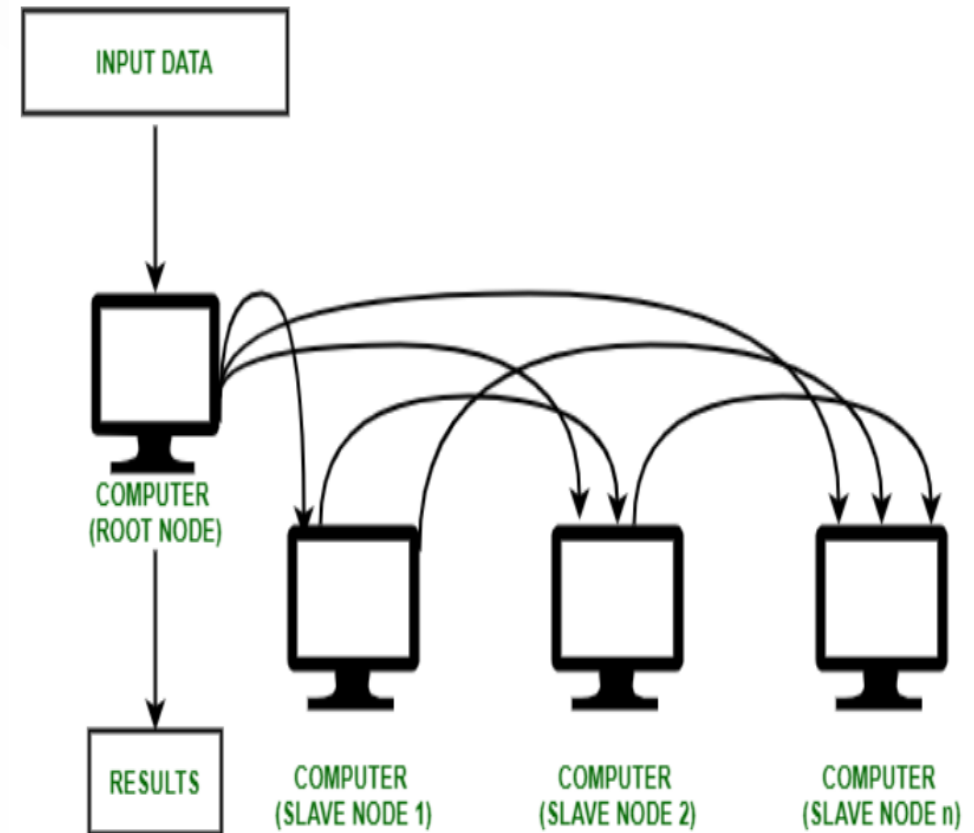
- **Client-server architectures.** The client-server architectures use smart clients that contact a server for data, then format and display that data to the user.
- **N-tier system architectures.** Typically used in application servers, these architectures use web applications to forward requests to other enterprise services.
- **Peer-to-peer architectures.** These divide all responsibilities among all peer computers, which can serve as clients or servers. This model is popular in several use cases, including blockchain networks, content exchange and media streaming.

Disadvantages of Distributed Computing

- **Configuration issues.** For a distributed system to work properly, all nodes in the network should have the same configuration and be able to interact with one another.
- **Communication overhead.** Coordinating and communicating between multiple nodes in a distributed system can create additional communication overhead, potentially reducing overall system performance.
- **Security management.** Managing security in distributed systems can be complex, as users should control replicated data across various locations and ensure network security.
- **Cost.** While distributed systems can be cost-effective in the long run, they often incur high deployment costs initially.

Cluster Computing

- Cluster computing is a kind of distributed computing, a type of computing that links computers together on a network to perform a computational task, increase computational power and function as a single computer.
- Each computer, or “node,” in a computer network has an operating system (OS) and a central processing unit (CPU) core that handles the tasks required for the software to run properly



A Simple Cluster Computing Layout

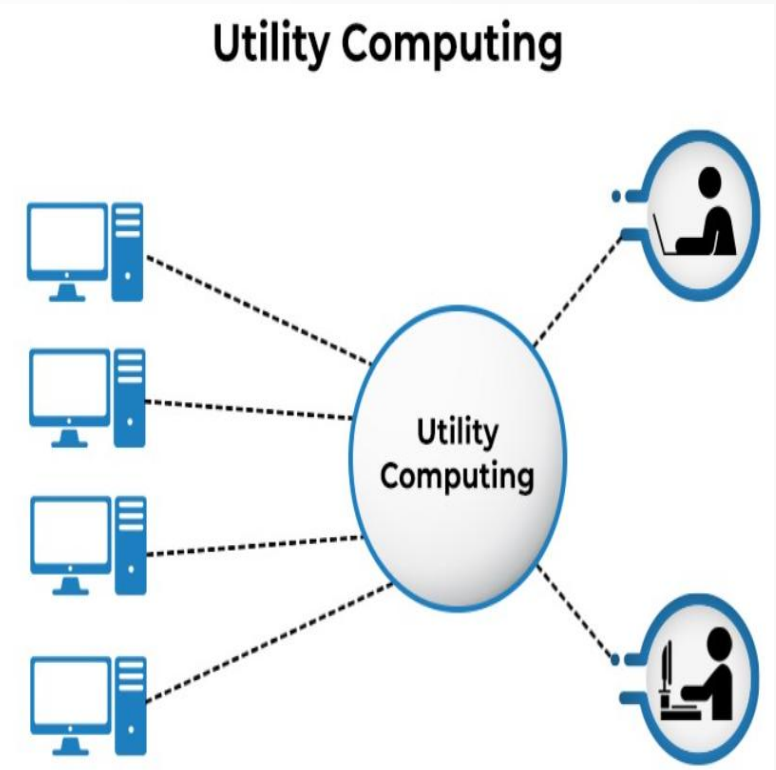
Cluster Computing

- At its most fundamental level, cluster computing uses a LAN to connect multiple, independent computers in a network.
- In the architecture of the cluster, each computer on the network is referred to as a “node” and is controlled by middleware, software that enables communication between each machine.
- Users of the cluster can use each computer's resources as though they were a single machine, rather than individual machines connected via a LAN.
- Because of its high performance and high availability, cluster computing has many applications, including cloud computing, artificial intelligence (AI), scientific research and big data analytics.

Cluster Computing	Grid Computing
Nodes must be homogeneous (same hardware and OS)	Nodes can be homogeneous or heterogeneous
Computers are dedicated to the same task	Computers contribute unused resources
Computers are located close to each other.	Computers may be located at a huge distance from one another.
Computers are connected by a high speed local area network bus.	Computers are connected using a low speed bus or the internet.
Computers are connected in a centralized network topology.	Computers are connected in a distributed or decentralized network topology.
Scheduling is controlled by a central server.	It may have servers, but mostly each node behaves independently.
Whole system has a centralized resource manager.	Every node manages it's resources independently.
Cluster computing is used in areas such as WebLogic Application Servers, Databases, etc.	Grid computing is used in areas such as predictive modeling, Automation, simulations, etc.
It has Centralized Resource management.	It has Distributed Resource Management.

Utility Computing

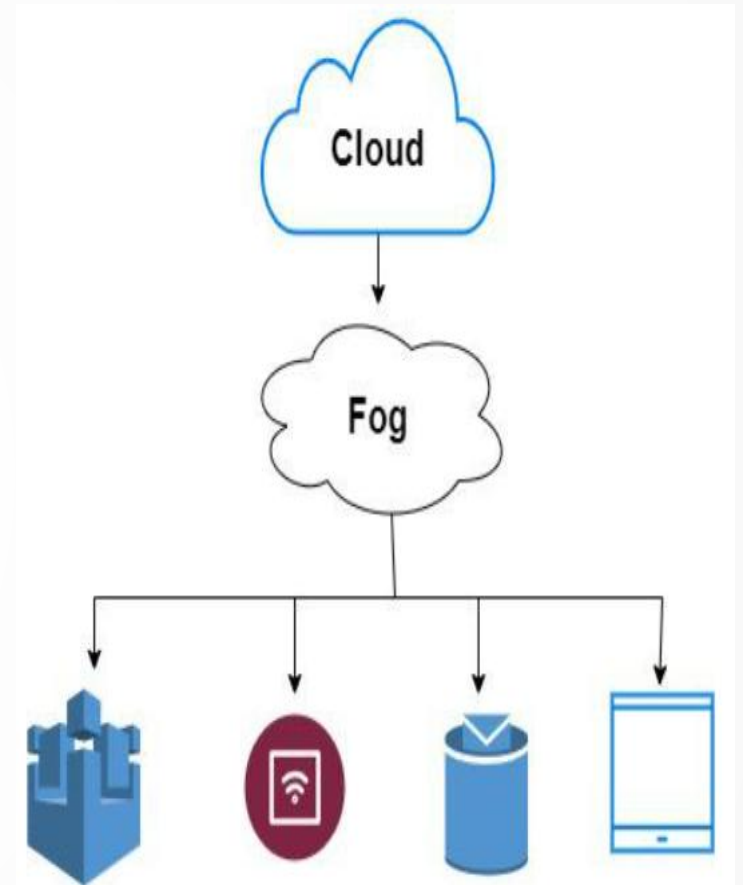
- Utility computing is defined as the type of computing where the service provider provides the needed resources and services to the customer and charges them depending on the usage of these resources as per requirement and demand, but not of a fixed rate.
- Utility computing involves the renting of resources such as hardware, software, etc. depending on the demand and the requirement.
- The goal of utility computing is to increase the usage of resources and be more cost-efficient.



Grid Computing	Utility Computing
It is a process architecture that combines different computing resources from multiple locations to achieve desired and common goal.	It is process architecture that provide on-demand computing resources and infrastructure on basis of pay per use method.
It makes better use of existing resources, address rapid fluctuations in customer demands, improve computational capabilities, provide flexibility, etc.	It simply reduces IT costs, easier to manage, provide greater flexibility, compatibility, provide more convenience, etc.
It mainly focuses on sharing computing resources.	It mainly focuses on acquiring computing resources.
It is used in ATMs, back-end infrastructures, marketing research, etc.	It is used in large organizations such as Amazon, Google, etc., where they establish their own utility services for computing storage and applications.
Its characteristics include resource coordination, transparent access, dependable access, etc.	Its characteristics include scalability, demand pricing, standardized utility computing services, automation, etc.

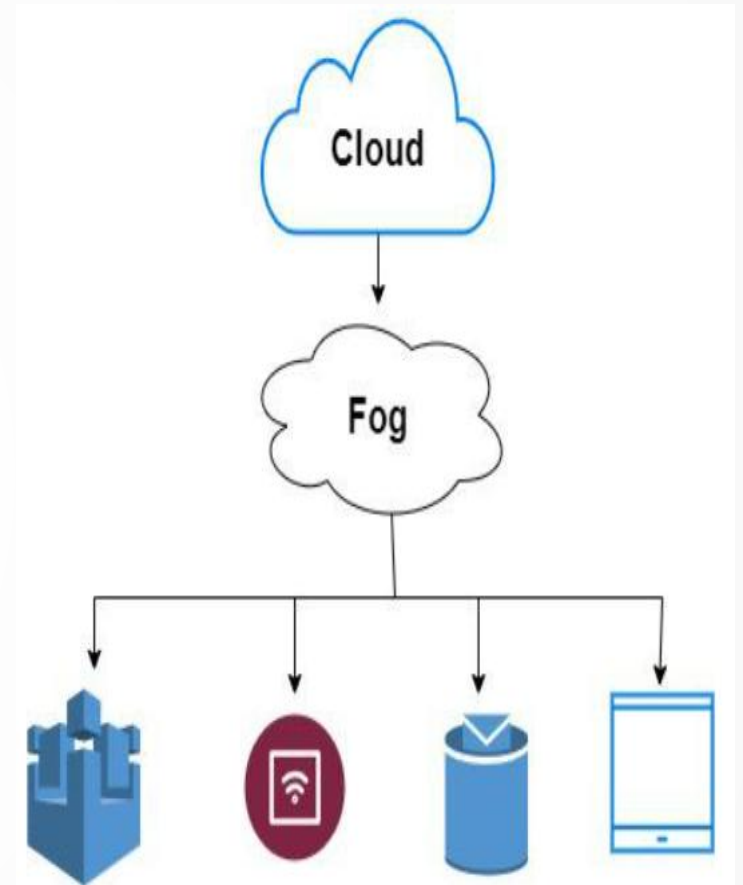
Fog Computing

- Fog Computing is introduced by Cisco that refers to **extending cloud computing to an edge of the enterprise's network**. Thus, it is also known as **Edge Computing** or **Fogging**.
- It facilitates the operation of computing, storage, and networking services between end devices and computing data centers.
- Fog computing is a distributed computing system that stores data and applications between the cloud and end-devices.
- Fog computing is used in areas that require low latency or real-time decisions, such as the Internet of Things (IoT), industrial automation, and autonomous vehicles.



Fog Computing

- The devices comprising the fog infrastructure are known as fog nodes.
- In fog computing, all the storage capabilities, computation capabilities, data along with the applications are placed between the cloud and the physical host.
- All these functionalities are placed more towards the host. This makes processing faster as it is done almost at the place where data is created.
- It improves the efficiency of the system and is also used to ensure increased security.



Why Is Fog Computing Used?

- **To improve latency and performance:** Because fog nodes are often deployed at the network edge, closer to the IoT devices themselves, they can substantially reduce the processing time and enhance performance for applications that demand low latency.
- **To improve decision-making:** It can help improve decision-making in real-time as fog computing allows for real-time data collection and analysis from IoT devices.
- **To reduce costs:** Fog computing can also help reduce costs associated with data storage and analysis. This is because, by bringing computation and data storage closer to the network edge, fog computing reduces the amount of data that needs to be transmitted back to a central location for processing.

Types of Fog Computing

1. **Device-level fog computing** runs on devices such as sensors, switches, routers, and other low-powered hardware. It can be used to gather data from these devices and send it to the cloud for analysis.
2. **Edge-level fog computing** runs on servers or appliances located at the edge of a network. These devices can be used to process data before it is sent to the cloud.
3. **Gateway-level fog computing** runs on devices that act as a gateway between the edge and the cloud. These devices can be used to manage traffic and ensure that only relevant data is sent to the cloud.
4. **Cloud-level fog computing** runs on servers or appliances located in the cloud. These devices can be used to process data before it is sent to end users.

Where Is Fog Computing Needed?

Fog computing is often used in cases where real-time response is needed, such as with industrial control systems, video surveillance, or autonomous vehicles. There are many potential applications for fog computing, including:

1. **Connected cars** — collecting and processing data from sensors in real-time to enable features such as autonomous driving and infotainment.
2. **Smart cities** — monitoring traffic flows, managing public transportation, optimizing energy use, and more.
3. **Industrial IoT** — enhancing efficiency and safety in factories, power plants, mines, and other industrial infrastructure.
4. **Connected health** — supporting remote patient monitoring, telemedicine, and other healthcare applications.
5. **AR/VR** — enabling low-latency, high-quality augmented and virtual reality experiences.

Cloud Computing

- Cloud computing is a model for delivering computing resources, like storage, servers, and software, over the internet.
- It is also referred to as Internet-based computing, that allows users to access and store information without having to manage their own physical devices or IT infrastructure
- Transparency, scalability, security and intelligent monitoring are some of the most important constraints which every cloud infrastructure should experience.
- Organizations of every type, size, and industry are using the cloud for a wide variety of use cases, such as data backup, disaster recovery, email, virtual desktops, software development and testing, big data analytics, and customer-facing web applications.

Cloud Computing

For example,

1. Healthcare companies are using the cloud to develop more personalized treatments for patients.
2. Financial services companies are using the cloud to power real-time fraud detection and prevention.
3. Video game makers are using the cloud to deliver online games to millions of players around the world.

Some cloud providers are Google cloud, AWS, Microsoft Azure and IBM cloud.

NIST: Cloud Computing Definition

- NIST (National Institute of Standards and Technology) is a U.S. government agency that develops standards and guidelines to improve the quality of products, services, and systems across various fields.
- NIST plays a crucial role in establishing standards and best practices for technology and cybersecurity, especially in areas such as cloud computing, cryptography, and data privacy.

NIST defines cloud computing as:

*"Cloud computing is a model for enabling **ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."*

NIST: Cloud Computing Definition

This definition highlights several core aspects of cloud computing, including:

1. **Ubiquitous, On-demand, and Convenient Network Access:** Cloud computing allows access from anywhere over the internet, making it convenient for users to access services and resources. Users can access and use cloud resources whenever they need them, and only pay for what they use
2. **Shared Pool of Configurable Resources:** Resources like servers, storage, and applications are shared and can be dynamically allocated and reallocated, ensuring efficient resource usage.
3. **Rapid Provisioning and Release:** Cloud resources can be quickly provisioned and de-provisioned, making the system highly flexible and adaptable to varying demands.
4. **Minimal Management Effort or Service Provider Interaction:** Cloud computing abstracts much of the complexity involved in resource management, so users don't need to manage infrastructure themselves.

NIST: Cloud Computing Characteristics

Five Essential Characteristics of Cloud Computing

1. **On-Demand Self-Service:** Users can provision resources like storage and compute power without requiring human interaction from the service provider.
2. **Broad Network Access:** Services are available over the network and can be accessed through various devices, including mobile phones, laptops, etc.
3. **Resource Pooling:** Cloud providers pool their resources to serve multiple consumers using multi-tenant models. Resources are dynamically assigned and reassigned according to consumer demand.
4. **Rapid Elasticity:** Resources can be rapidly scaled up or down to match demand, providing flexibility and efficiency.
5. **Measured Service:** Cloud resources are metered, and users are billed according to their usage, ensuring cost-effectiveness.

Cloud Computing Service Models

Cloud computing offers the following three type of services:

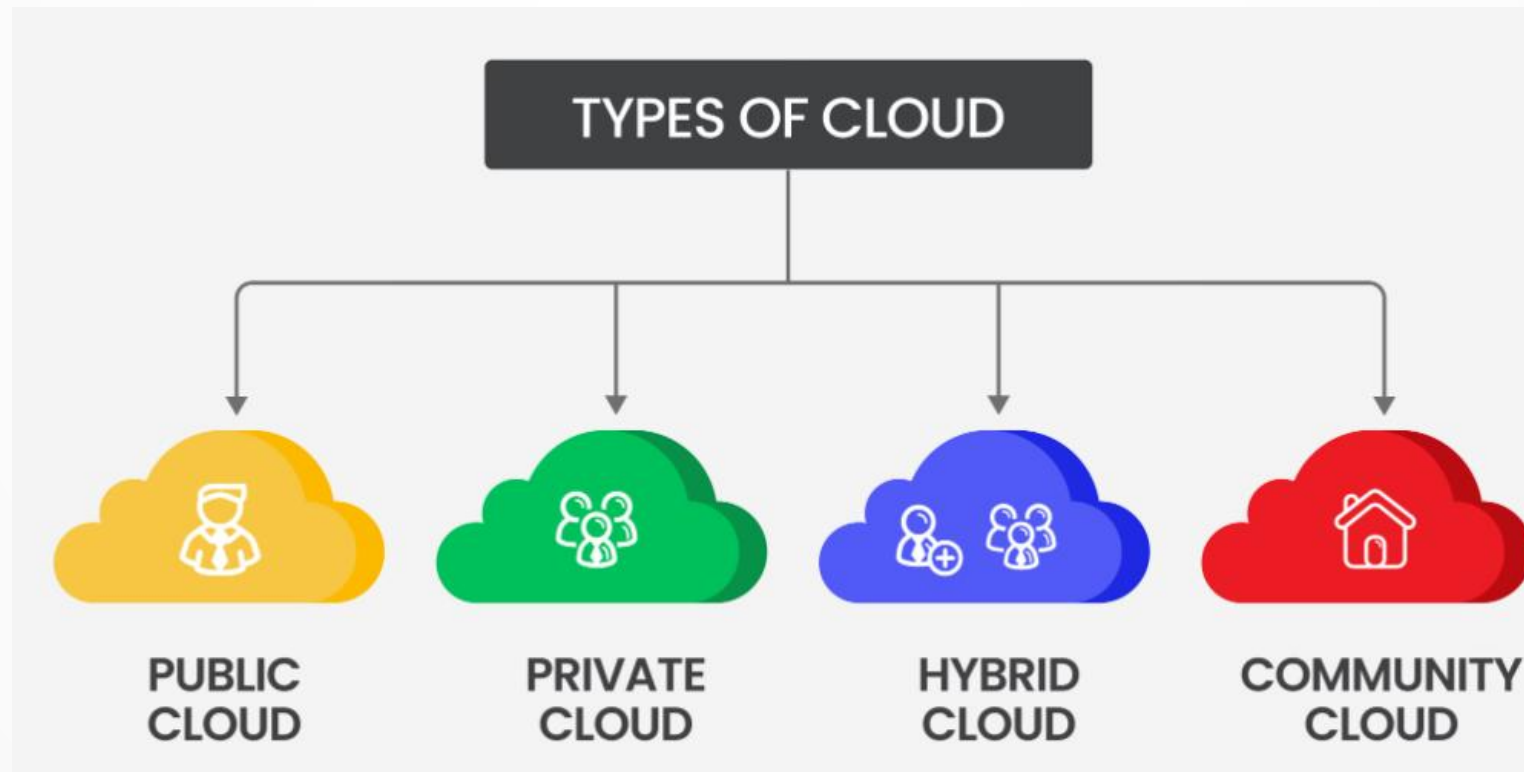
Software as a Service (SaaS) – It is also known as cloud application services. Mostly, SaaS applications run directly through the web browser means we do not require to download and install these applications. Example: Google Apps, Salesforce Dropbox, Slack, Hubspot, Cisco WebEx.

Platform as a Service (PaaS) – It is also known as cloud platform services. It is quite similar to SaaS, but the difference is that PaaS provides a platform for software creation, but using SaaS, we can access software over the internet without the need of any platform. Example: Windows Azure, Force.com, Magento Commerce Cloud, OpenShift.

Infrastructure as a Service (IaaS) – It is also known as cloud infrastructure services. It is responsible for managing applications data, middleware, and runtime environments. Example: Amazon Web Services (AWS) EC2, Google Compute Engine (GCE), Cisco Metapod.

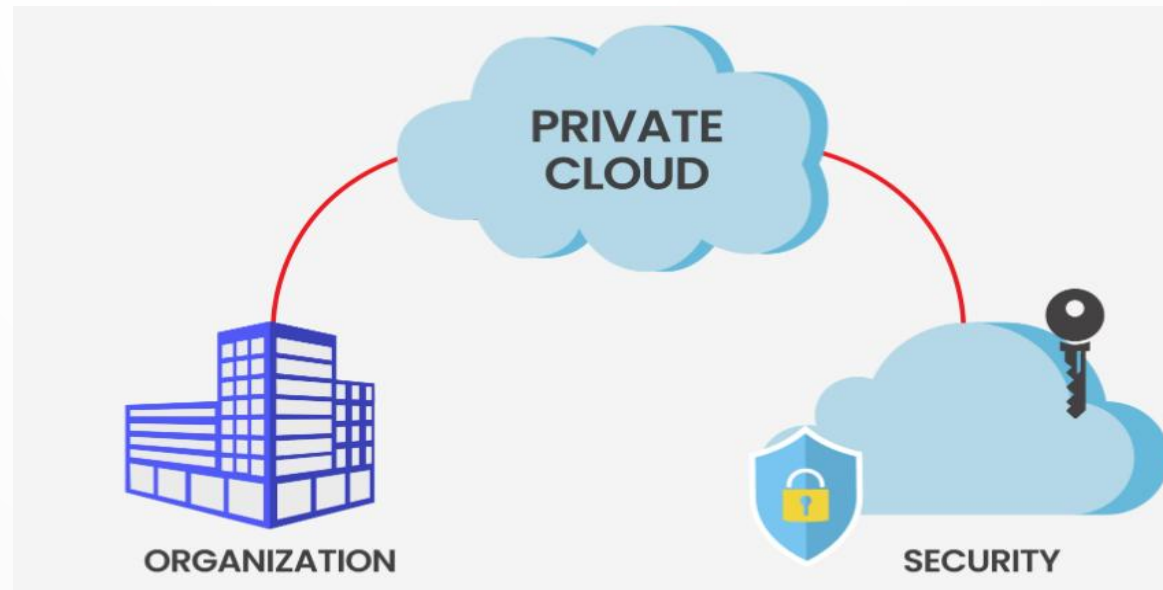
Cloud Computing Deployment Models

- A cloud deployment model is a **conceptual framework that describes the relationship between the infrastructure and the end-user.**
- It is often defined as the place where the deployment will take place.
- Each cloud deployment model has its own unique characteristics, and it is important that you select one that fits your organization's needs.



Private Cloud

- The cloud infrastructure is provisioned for **exclusive use by a single organization** comprising multiple consumers (e.g., business units).
- A company can choose to have its own private cloud platform or run it on-premise.
- It may be owned, managed, and operated by the organization, a third party, or some combination of them. This ensures that the systems are designed and operated according to the company's specifications.



Private Cloud

Private cloud deployments are used for medical companies, banking institutions, and organizations with state guidelines for data use control.

Pros:

1. **Control.** Fewer people having access to the admin tools and configuration of your private cloud will give you more control.
2. **Customization.** If you have a compelling reason to develop a new feature, you can easily deploy it in-house.
3. **Security.** You can easily integrate as many security services as your business needs. Two-factor Authentication is easily the most secure method of cloud security.

Cons:

1. **Team.** To take advantage of the power of the private cloud, you need the right skills of developers, security experts, DevOps and etc.
2. **Cost.** Most companies can't afford to set up their own cloud infrastructure. Even with the most advanced technology, the cost of running a private cloud is still expensive.

Community cloud:

- The cloud infrastructure is provisioned for **exclusive use by a specific community of consumers** from organizations that have shared concerns.
- This deployment model supports multi-organizations sharing a common cloud environment.
- For example, universities share computing resources with the police force. Access to a community cloud environment is typically restricted to unauthorized members.
- It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.





Community cloud:

Pros:

Cost. It is cost-effective because of sharing the cloud with several organizations or communities.

Collaboration. It is suitable to share resources, infrastructure, and other features among various organizations and members.

Cons:

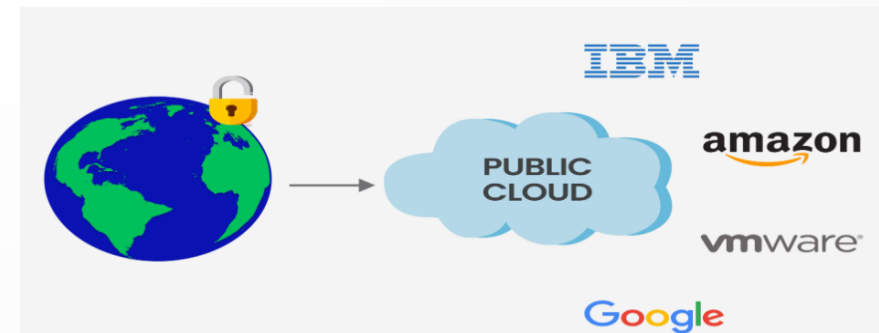
Limitations. There is a fixed amount of data storage is shared among all members. Also, not every organization is suitable to use it.

Security. Security features are better than public cloud but are not good as private capabilities.

Public cloud

The cloud infrastructure is provisioned for open use by the general public.

- Public cloud deployments are usually housed on public servers that are accessible over the internet.
- The hardware and software used in these facilities are typically owned by the service owner. This saves a huge amount of money for companies that do not want to buy their own application server, storage, CPU, OS, database and etc.
- This model eliminates the need for additional staff members to maintain and service components. It also allows you to easily expand and add new services without having to hire more people.
- Most of all, public clouds are used for file-sharing, e-mail services, application development, and testings.



Public cloud

Pros:

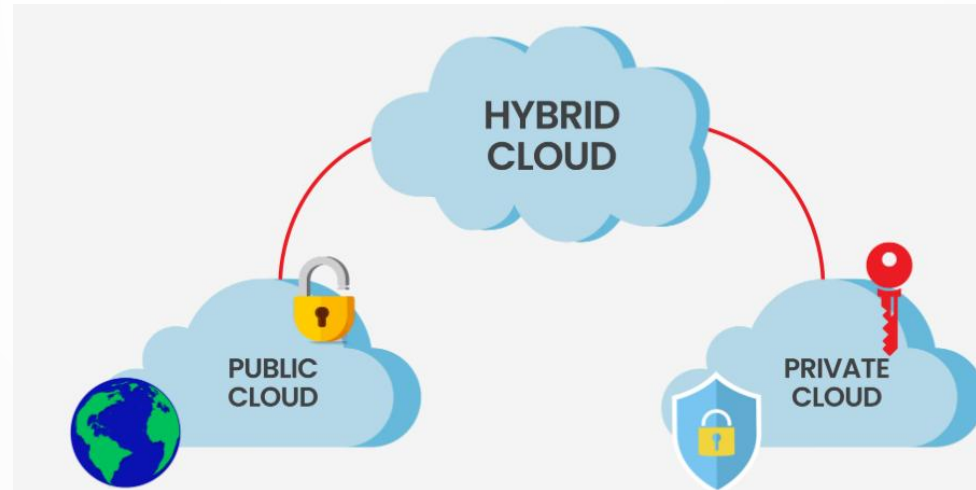
- **Easy to manage.** It eliminates the need for you and your team to regularly attend to the maintenance of the system.
- **Cost.** The way services are billed is that you pay for what you use, and then regulate when it needed.
- **Reliability and performance.** There is no limit to the number of users. And the location of users is independent because its services are delivered through the internet.

Cons:

- **Control.** In the event of a failed cloud platform, you would not be able to maintain continuity of operations.
- **Security.** Your data is shared for big amount of users.

Hybrid cloud

- Hybrid cloud deployment models are often formed by taking the best of both worlds – private and public cloud deployment models.
- They combine various aspects of a company's requirements into a single, unified cloud solution.
- For example, the private cloud can be used to store sensitive data, while the public cloud can be used to store user functions.



Hybrid cloud

Pros:

Flexibility. One of the best features of the cloud type is that it's very flexible. You can select the parts of it that you would like to integrate into your solution.

Scalability. You can regulate the functionality with the demand of users.

Cons:

Data silos. They are the places where all of your collected data lies. Having the proper separation of these silos can help ensure that all of your data is properly allocated.

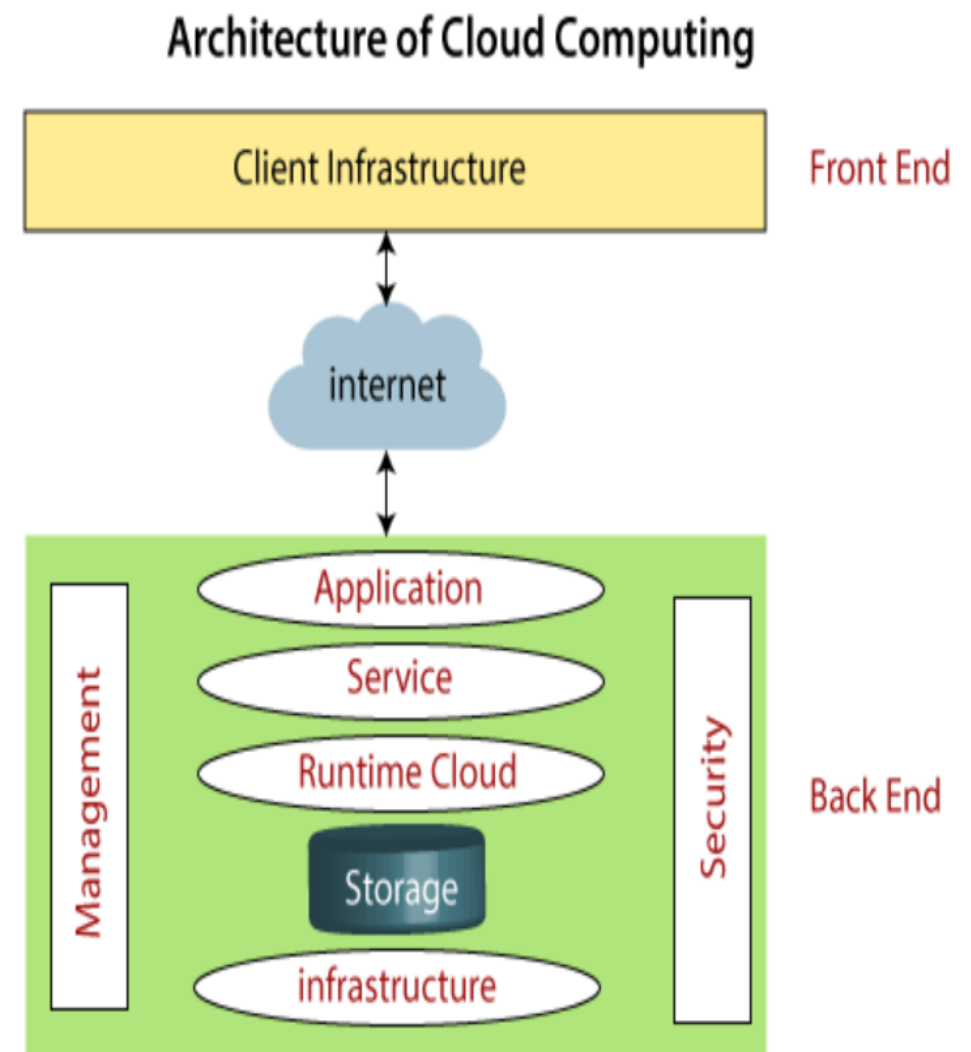
Cost. Hybrid cloud delivery models are not as expensive as they may seem, but they are also prone to spending too much. There is a risk of spending too much if you choose incorrect cloud services.

Comparison of 4 cloud deployment models

PARAMETERS	PUBLIC	PRIVATE	HYBRID	COMMUNITY
SCALABILITY	Very High	Limited	Very High	Limited
HOST	Service provider	Thrid party	Third party	Community
RELIABILITY	Moderate	Very High	Medium to High	Very High
USERS	General public	Selected users	Selected users	Community members
SECURITY	Depends on service provider	High Class Security	Secure	Secure
PERFORMANCE	Low to Medium	Good	Good	Very Good
COST	Cheaper	High Cost	Costly	Costly
EXAMPLES	Amazon EC2	Microsoft	IBM	VMWare

Cloud Computing Architecture

- Cloud computing architecture is a combination of service-oriented architecture and event-driven architecture.
- Client infrastructure, application, service, runtime cloud, storage, infrastructure, management and security all these are the components of cloud computing architecture.
- Cloud computing architecture is divided into the following two parts -
 - Front End
 - Back End



Cloud Computing Architecture

Frontend

- The front end is used by the client.
- It contains client-side interfaces and applications that are required to access the cloud computing platforms.
- The front end includes web servers (including Chrome, Firefox, internet explorer, etc.), thin & fat clients, tablets, and mobile devices.

Backend

- Backend refers to the cloud itself which is used by the service provider. It contains the resources as well as manages the resources and provides security mechanisms.
- Along with this, it includes huge storage, virtual applications, virtual machines, traffic control mechanisms, deployment models, etc.

Both front end and back end are connected to others through a network, generally using the internet connection.

Components of Cloud Computing Architecture

There are the following components of cloud computing architecture

1. **Client Infrastructure:** Client Infrastructure is a Front end component. It provides GUI (Graphical User Interface) to interact with the cloud.
2. **Application:** The application may be any software or platform that a client wants to access.
3. **Services:** A Cloud Services manages that which type of service you access according to the client's requirement.
4. **Runtime Cloud:** Runtime Cloud provides the **execution and runtime environment** to the virtual machines.
5. **Storage:** Storage is one of the most important components of cloud computing. It provides a huge amount of storage capacity in the cloud to store and manage data.

Components of Cloud Computing Architecture

6. **Storage:** Storage is one of the most important components of cloud computing. It provides a huge amount of storage capacity in the cloud to store and manage data.
7. **Infrastructure:** It provides services on the **host level**, **application level**, and **network level**. Cloud infrastructure includes hardware and software components such as servers, storage, network devices, virtualization software, and other storage resources that are needed to support the cloud computing model.
8. **Management:** Management is used to manage components such as application, service, runtime cloud, storage, infrastructure, and other security issues in the backend and establish coordination between them.
9. **Security:** Security is an in-built back end component of cloud computing. It implements a security mechanism in the back end.
10. **Internet:** The Internet is medium through which front end and back end can interact and communicate with each other.

Need of Green Cloud Computing

Traditional servers, commonly used in on-premises data centers, contribute significantly to carbon emissions due to:

1. **High Energy Consumption:** Servers require continuous power to operate, consuming a large amount of electricity. Most electricity is still generated from fossil fuels, leading to carbon emissions.
2. **Cooling Systems:** Data centers rely heavily on air conditioning systems to keep servers cool, which adds to energy usage.
3. **Inefficiency:** Traditional servers are often underutilized, running at only 10–30% capacity while still consuming significant power.
4. **Non-renewable Energy:** Many traditional data centers rely on non-renewable energy sources, such as coal and natural gas, further increasing their carbon footprint.

Need of Green Cloud Computing

- Green cloud computing is essential due to the **significant environmental impact of traditional data centers** and the increasing energy demands of modern technologies like artificial intelligence (AI).
- Data centers worldwide **consume vast amounts of electricity, contributing to carbon emissions and environmental degradation.**
- For instance, by 2025, data centers are projected to consume 20% of global electricity and emit up to 5.5% of the world's carbon emissions.
- The rapid growth of AI and advanced computing has further escalated energy consumption.
- To mitigate these environmental challenges, green cloud computing focuses on strategies such as server virtualization, adoption of renewable energy sources, and energy-efficient infrastructure designs.

Green Cloud Computing

'Green Cloud' refers to an **environment-friendly cloud that reduces carbon emissions and promotes the use of renewable energy to reduce energy needs.**

It is a **set of beneficial practices** which reduce energy consumption and carbon emissions.

Green computing advocates the use of renewable energy from different sources.

It has three main goals:

- **Maximizing energy efficiency**
- **Promoting recycling**
- **Minimizing the use of hazardous materials**

All the major cloud providers have taken steps to convert their data centers into green cloud. **AWS's sustainability pillar** is one example of how Amazon promotes an environment-friendly cloud.

Green Cloud Computing

AWS's sustainability pillar is one example of how Amazon promotes an environment-friendly cloud.

AWS's Commitment to Sustainability

- **Carbon Footprint Reduction:** AWS is working towards achieving net-zero carbon emissions by 2040.
- **Renewable Energy:** AWS aims to power its global operations with 100% renewable energy by 2025.
- **Energy Efficiency:** AWS data centers are designed for energy efficiency, using advanced cooling techniques and energy-efficient hardware.

Green Cloud Computing

Key Principles

1. **Energy Efficiency:** Designing data centers to consume less power and operate more efficiently
2. **Renewable Energy:** Using solar, wind, and other renewable energy sources to power cloud infrastructure
3. **Carbon Offsetting:** Implementing measures to offset carbon emissions generated by cloud operations

Benefits of Cloud Computing:

1. **Reduced Carbon Footprint:** Green cloud computing significantly lowers greenhouse gas emissions
2. **Alignment with Corporate Sustainability Goals:** Supports businesses in achieving their environmental objectives
3. **Long-Term Cost Savings:** Reduces operational costs through energy-efficient practices and renewable energy usage

Sustainable Practices for Green Cloud Computing

- **Energy-Efficient Data Centers:** Optimize data centers by using **energy-efficient hardware, advanced cooling systems, and power management techniques** to reduce energy consumption and promote green cloud computing sustainability.
- **Renewable Energy Usage in Cloud Solutions:** Incorporate **renewable energy sources** such as solar and wind power into your cloud infrastructure. Many cloud providers offer **options** to use renewable energy data centers.
- **Sustainable Software Design:** Developing software that requires less computational power and can run efficiently on lower-end hardware.
- **E-Waste Management:** Implementing proper disposal and recycling methods for outdated or obsolete hardware to minimize electronic waste.
- **Green Policies and Governance:** Establishing policies that promote sustainable practices within organizations and among stakeholders.

Sustainable Practices for Green Cloud Computing

- **Carbon-Neutral Cloud Computing:** Implement carbon offsetting measures such as **purchasing carbon credits or investing in projects** that reduce greenhouse gas emissions to compensate for the carbon footprint of your cloud operations, achieving carbon-neutral cloud computing.
 - **Carbon Credits:** These are permits or certificates that allow an organization to emit a specific amount of CO₂. Companies can purchase carbon credits to offset their emissions by investing in environmental projects like reforestation or renewable energy development.

Example: A cloud provider calculates that their data center emits 1,000 tons of CO₂ annually. They purchase carbon credits that fund a wind farm capable of reducing emissions by the same amount.
 - **Investing in Green Projects:** Organizations can directly invest in projects aimed at reducing greenhouse gases, such as solar or wind energy projects, methane capture initiatives, or forest conservation.

Example: Amazon Web Services (AWS) invests in renewable energy projects like wind farms in the US, which supply clean energy to power their data centers.

Five Ideas of Green Computing

- **Virtualization:** Using virtualization to maximize resource utilization and reduce the number of physical servers required
- **Efficient Data Centers:** Designing data centers with energy-efficient hardware, advanced cooling systems, and power management techniques
- **Renewable Energy:** Incorporating renewable energy sources like solar and wind power to run data centers and cloud infrastructure
- **Energy Audits:** Conducting regular energy audits to identify areas for improvement and reduce energy consumption
- **Carbon Offsetting:** Implementing carbon offsetting measures such as purchasing carbon credits or investing in projects that reduce greenhouse gas emissions

Example of Green Computing

- **Google:** Google has achieved carbon neutrality and aims to operate on 24/7 carbon-free energy by 2030.

By using renewable energy and optimizing data center efficiency, Google has significantly reduced its carbon footprint

- **Microsoft** has set a bold **goal to become carbon negative by 2030** and remove all historical emissions by 2050.

Their strategy includes reducing emissions, investing in new carbon removal technologies, and supporting global carbon reduction efforts.

Microsoft plans to fund carbon removal technologies, help suppliers cut their carbon footprints, and enhance transparency through annual environmental reports.

Example of Green Computing

- **AWS** is also dedicated to sustainability and has, thus far, achieved 100% renewable electricity usage seven years ahead of its 2030 goal.
- The company focuses on maximizing energy efficiency through advanced data center design, cooling innovations, and power management technologies.
- AWS's infrastructure is reported to be up to 4.1 times more efficient than traditional data centers, with their Graviton4 and Inferentia2 chips contributing to enhanced performance and energy efficiency.

Cloud Computing Service Model: IaaS

- **Infrastructure as a Service (IaaS)** is a cloud computing model that provides **virtualized computing resources** over the internet.
- It offers **fundamental infrastructure components such as virtual machines (VMs), storage, networking, and operating systems on a pay-as-you-go basis.** This eliminates the need for businesses to invest in and maintain physical hardware.
- **Key Features of IaaS**
 1. **Virtualized Computing Resources**
 1. Provides virtual machines (VMs) with scalable computing power.
 2. Users can configure CPU, RAM, and disk storage based on their requirements.
 2. **Storage and Backup Solutions**
 1. Cloud providers offer scalable storage options (e.g., Amazon S3, Google Cloud Storage).
 2. Backup and disaster recovery features ensure data safety.

Cloud Computing Service Model: IaaS

■ Key Features of IaaS

3. On-Demand Scalability

1. Infrastructure can be expanded or reduced based on usage.
2. Helps in handling sudden workload spikes without investing in additional hardware.

4. Pay-as-You-Go Pricing

1. Users pay only for the resources they consume.
2. Eliminates capital expenditure (CAPEX) on IT infrastructure.

Cloud Computing Service Model: IaaS

IaaS is considered the lowest level because it serves as the foundation on which PaaS and SaaS are built.

It provides **raw infrastructure**, while PaaS and SaaS offer more abstraction and user-friendly environments.

► Examples of IaaS Providers

1. Amazon Web Services (AWS) – EC2 (Elastic Compute Cloud)
2. Microsoft Azure – Azure Virtual Machines
3. Google Cloud Platform (GCP) – Compute Engine
4. IBM Cloud – Virtual Servers
5. Oracle Cloud Infrastructure (OCI)

Cloud Computing Service Model: PaaS

- **Platform as a Service (PaaS)** is a cloud computing model that provides a **pre-configured development and deployment environment for building, testing, and managing applications.**
- It offers infrastructure, runtime environments, databases, and development tools **without requiring users to manage the underlying hardware or operating system.**
- PaaS enables developers to focus on coding and application logic rather than infrastructure management.

Examples of PaaS Providers

1. **AWS Elastic Beanstalk** – Deploy and scale applications easily.
2. **Google App Engine** – Serverless application hosting.
3. **Microsoft Azure App Services** – Cloud-based application hosting and management.
4. **IBM Cloud Foundry** – Enterprise-level PaaS for application development.
5. **Heroku** – Popular PaaS for developers using Ruby, Python, Java, and Node.js.

Cloud Computing Service Model: PaaS

Key Features of PaaS

1. Development and Runtime Environment

- Provides programming languages, frameworks, and runtime environments.
- Examples: Node.js, Java, .NET, Python, Ruby on Rails.

2. Pre-Configured Infrastructure

- Includes servers, storage, networking, and databases.
- No need to configure or manage operating systems and middleware.

3. Database Management

- Built-in databases like **AWS RDS**, **Google Cloud SQL**, **Azure SQL Database** help manage structured and unstructured data.

Cloud Computing Service Model: PaaS

Use Cases of PaaS

1. **Application Development & Hosting** – Quick deployment of web and mobile apps.
2. **Big Data Analytics** – Processing large datasets using AI/ML tools.
3. **Internet of Things (IoT) Applications** – Building IoT platforms with real-time data processing.
4. **Business Process Automation** – Automating workflows and enterprise applications.

PaaS is ideal for developers who want to build and deploy applications without managing hardware and OS layers.

Cloud Computing Service Model: SaaS

- **Software as a Service (SaaS)** is a cloud computing model that **provides fully managed software applications over the internet**. Users can access SaaS applications through a **web browser without needing to install, maintain, or update the software on their local devices**.
- SaaS eliminates the need for infrastructure management, making it the easiest-to-use cloud service model.
- **SaaS is at the highest level because users do not manage any infrastructure or platform components. They only interact with the software interface.**

Key Features of SaaS

1. On-Demand Software Access

- Applications are accessible **anytime, anywhere** via the internet.
- No need for local installation; just log in and start using.

Cloud Computing Service Model: SaaS

Key Features of SaaS

2. Automatic Updates & Maintenance

- Software providers handle updates, security patches, and maintenance.
- Ensures users always have the latest features without manual upgrades.

3. Pay-As-You-Go Pricing Model

- Users pay for only what they use (subscription-based pricing).
- Reduces upfront costs for businesses.

4. Security & Compliance

- Cloud providers handle **data security, backups, encryption, and compliance (GDPR, HIPAA, ISO 27001, etc.)**.

Cloud Computing Service Model: SaaS

5. Integration with Other Services

- SaaS applications integrate with third-party tools (e.g., CRM, email, analytics).
- APIs allow easy connectivity with other cloud services.

Examples of SaaS Providers

1. **Google Workspace (Gmail, Drive, Docs, Meet)** – Cloud-based productivity tools.
2. **Microsoft 365 (Word, Excel, Outlook, Teams)** – Business collaboration and office suite.
3. **Salesforce** – Cloud-based CRM software.
4. **Dropbox & Google Drive** – Cloud storage solutions.
5. **Zoom & Slack** – Video conferencing and messaging platforms.

Cloud Computing Service Model: SaaS

Use Cases of SaaS

- **Email & Collaboration** – Google Workspace, Microsoft 365.
- **Customer Relationship Management (CRM)** – Salesforce, HubSpot.
- **Enterprise Resource Planning (ERP)** – SAP, Oracle Cloud.
- **E-commerce & Online Sales** – Shopify, BigCommerce.
- **Cloud Storage & Backup** – Dropbox, Google Drive, OneDrive.

SaaS is ideal for businesses and individuals looking for a plug-and-play solution without managing any infrastructure or software updates.

Cloud Computing Service Model: BPaaS

- **Business Process as a Service (BPaaS)** is a cloud-based service model that provides **business process outsourcing (BPO)** solutions over the internet.
- It enables organizations to **automate and manage their business processes (such as payroll, HR, accounting, and customer service) without investing in IT infrastructure or hiring in-house experts.**
- BPaaS leverages cloud computing, automation, and AI to optimize business workflows, reducing costs and improving efficiency.
- The pricing is calculated according to the use of the service.
- Providers of BPaaS include IBM, Wipro, Ovum and Cognizant.

Cloud Computing Service Model: BPaaS

Why is BPaaS Important?

1. Helps businesses **focus on core competencies** while outsourcing repetitive tasks.
2. Reduces operational costs by **eliminating IT infrastructure and staffing expenses**.
3. Increases efficiency with **automation and AI-driven decision-making**.
4. Provides **data-driven insights** to improve business strategies.

Use Cases of BPaaS

Human Resource Management (HRM) – Payroll, recruitment, benefits administration.

Customer Service & Support – AI chatbots, call center outsourcing.

Finance & Accounting – Invoice processing, bookkeeping.

Supply Chain Management – Logistics tracking, vendor coordination.

Marketing & Sales Automation – Lead generation, email marketing campaigns.

Feature	BPaaS	SaaS	PaaS	IaaS
Focus	Business Processes	Software Applications	App Development	IT Infrastructure
Who Manages It?	Provider (Automated/AI-driven)	Provider	Provider	Provider manages infrastructure, user manages OS & apps
Customization	High	Low	Medium	High
Example Services	Payroll, HR, CRM, Accounting	Gmail, Salesforce, Office 365	Google App Engine, AWS Lambda	AWS EC2, Google Compute Engine

Cloud Computing Service Model: HaaS

Hardware as a Service (HaaS) is a business model where **hardware resources (such as servers, storage devices, network equipment, or computers)** are provided as a service rather than purchased outright.

This model allows organizations to **access and use hardware infrastructure on a subscription or rental basis**, rather than making a large capital investment in physical hardware.

Key Features of HaaS:

- 1. Cost Efficiency:** With HaaS, businesses only pay for the hardware they use, eliminating the need for large upfront costs. This is often more affordable, especially for smaller businesses or startups.
- 2. Scalability:** As your needs grow or change, you can easily scale your hardware usage up or down. This flexibility makes it easier to adjust to fluctuating demands without worrying about hardware obsolescence.

Cloud Computing Service Model: HaaS

Key Features of HaaS:

- 3. Maintenance and Support:** In a HaaS model, the service provider typically handles maintenance, updates, and troubleshooting. This reduces the burden on IT teams within the organization.
- 4. Up-to-date Technology:** HaaS allows businesses to use the latest hardware without needing to upgrade their systems frequently. This is particularly beneficial in industries where staying current with technology is crucial.
- 5. Pay-As-You-Go:** Instead of purchasing expensive hardware, businesses can pay for the hardware on a subscription basis, which may include an additional layer of service like cloud computing, software updates, and technical support.

Cloud Computing Service Model: HaaS

Example of HaaS:

1. A company could rent **servers or network equipment that are housed in a data center**. The service provider ensures that the hardware is up to date, performs necessary maintenance, and provides support if something goes wrong.
2. Another example is a business leasing computer systems for its employees. The service provider manages updates, repairs, and replacements, reducing the company's overhead.

Cloud Computing Service Model: HaaS

Advantages of HaaS:

1. **Cost predictability:** Fixed monthly costs for hardware usage make budgeting easier.
2. **No hardware depreciation:** Businesses don't have to worry about the depreciation of their hardware.
3. **Focus on core business:** With less focus on hardware maintenance and upgrades, organizations can focus more on their core operations.

Disadvantages of HaaS:

1. **Dependency on the provider:** If the provider experiences outages or issues, it can impact the business.
2. **Limited customization:** Sometimes, the hardware available as a service might not be as customizable as hardware owned outright.

Cloud Computing Service Model: HaaS

- **Hardware as a Service (HaaS)** refers to the **renting or leasing of computing infrastructure from a cloud service provider**. This could include **servers, storage, and networking hardware**, as well as other equipment needed for cloud services.
- HaaS providers handle hardware installation, maintenance, upgrades, and decommissioning, as well as providing the necessary infrastructure to keep it running, which may also include power, cooling systems, and data center space. The hardware is generally hosted in the service provider's data center and customers access the resources over the internet.
- This model allows organizations to reduce the costs and responsibilities associated with owning, maintaining, and housing their own hardware. Furthermore, since the hardware is managed remotely, organizations can scale their hardware needs up or down easily and quickly in response to their needs.

HaaS vs. IaaS

Hardware as a Service and Infrastructure as a Service (IaaS) are both models that provide essential IT resources through a service-based approach, but they differ in scope and application.

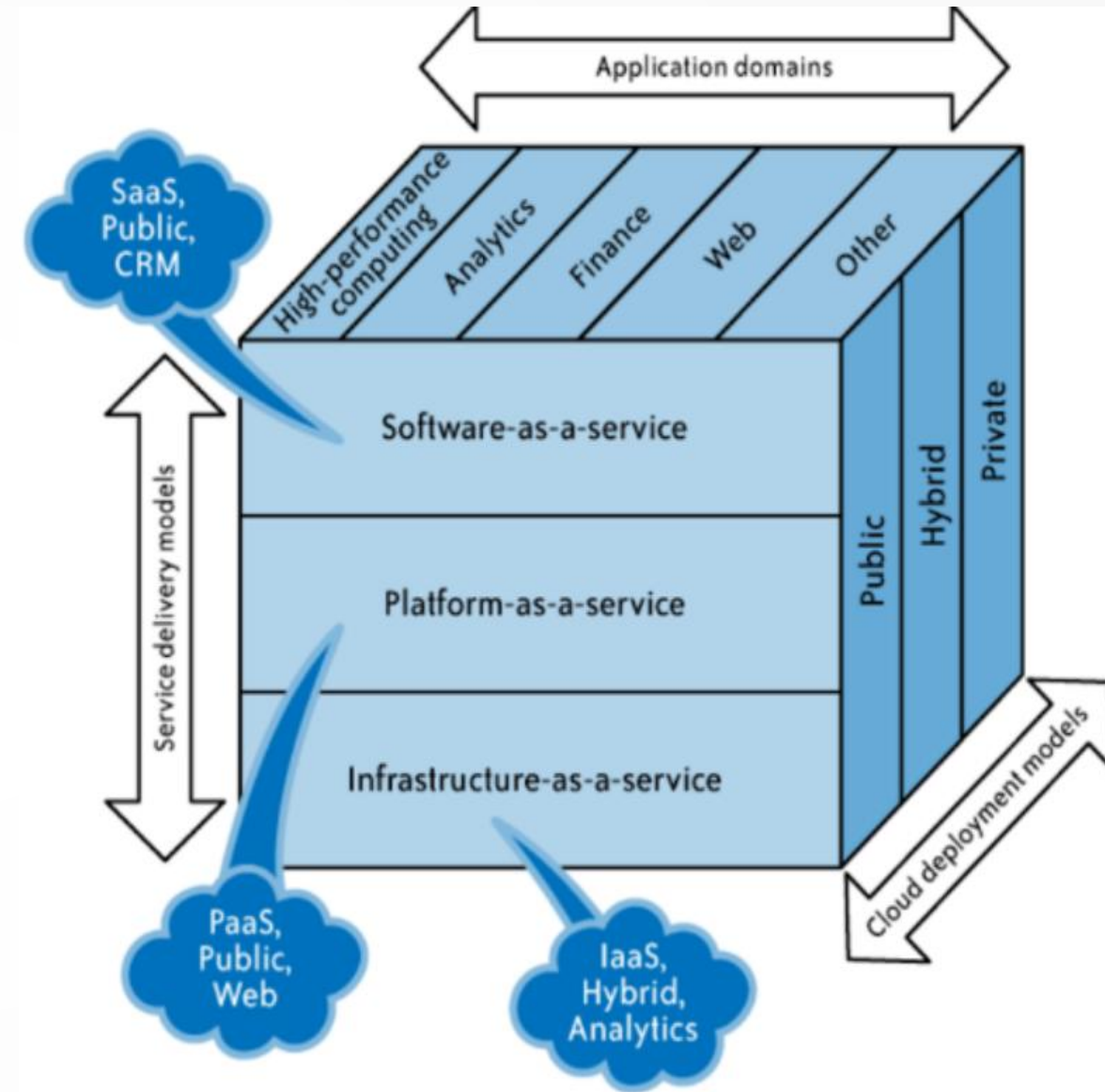
HaaS	IaaS
HaaS focuses specifically on the provision of physical hardware , such as network devices, servers, and storage systems, which are managed and maintained by the service provider.	IaaS offers a broader range of services that not only include the hardware but also encompass the virtualized infrastructure components like virtual machines, networking, and storage capabilities, typically hosted in the cloud
HaaS is more hardware-centric	IaaS provides a more comprehensive set of services that include both hardware and virtual infrastructure management.

SPI model

The SPI model is an abbreviation of the most common cloud computing service models: software as a service (SaaS), platform as a service (PaaS) and infrastructure as a service (IaaS).

The Figure shows the relationship between services, uses, and types of clouds.

Based on its business goals, an enterprise can choose to adopt one, or multiple, of the cloud service types depicted in the SPI model: SaaS, PaaS, IaaS.



SPI model

In general, all three of these cloud service models follow a **pay-as-you-go approach** and reduce the hardware and software investments an enterprise needs to make within its own on-premises data center.

The three types of cloud computing service models within the SPI framework can be deployed as part of a public cloud, private cloud, hybrid cloud or multi-cloud strategy.

What is Traditional Computing!!

■ **Traditional Computing: Your Own Equipment**

- Traditional computing refers to the use of electronic devices such as desktop and laptop computers, that use a central processing unit (CPU) to execute instructions stored in memory.
- Traditional computing, also known as **on-premises computing**, means you have physical servers and hardware at your workplace.
- You buy and maintain this equipment, and you need IT experts to set it up and keep it running.

Traditional Computing Pros

1. **Reliability:** Traditional computers have been around for decades and are **well understood by computer technicians and engineers**, making them easy to repair and maintain if something goes wrong.

Additionally, traditional computers tend to have **fewer compatibility issues than newer models** since they use older hardware and software that is better supported by manufacturers and developers.

2. **Performance:** Traditional computing also offers superior performance compared to their more modern counterparts. Traditional desktop PCs often come with more **powerful processors, larger amounts of RAM, and higher-end graphics cards** than laptops or tablets, which allows them to handle demanding tasks such as gaming or video editing with ease.

Additionally, traditional desktops can often be easily upgraded with additional components when needed in order to keep up with changing technology demands.

Traditional Computing Pros

3. **Customizability:** Traditional desktops also offer more customizability than other types of computers due to their modular design. **For example, users can easily swap out components such as the processor or graphics card in order to upgrade their system's performance without having to purchase an entirely new computer.**

This makes it easy for users to tailor their system's hardware configuration specifically to their needs without having to worry about compatibility issues that may arise from mixing different components together from different manufacturers.

4. **Ease Of Use:** Traditional desktop PCs offer a user-friendly experience with their straightforward design and intuitive user interface.

Unlike laptops or tablets that rely on touch gestures to access certain features, traditional desktop PCs provide the convenience of physical buttons, making them easier to navigate and operate.

Traditional Computing Cons

1. **High Cost:** Traditional computing is often **more expensive** than other forms of computing, such as cloud computing or edge computing.

This is because traditional computing requires the purchase of hardware, such as servers and storage systems, which can be costly to purchase and maintain.

Additionally, traditional computing also requires specialized software licenses, which can add to the cost.

2. **Complex Infrastructure:** Setting up a traditional computing environment can be complex and time-consuming.

It requires setting up physical infrastructure, such as servers and storage systems, as well as networking equipment and software applications.

This process can take weeks or even months to complete depending on the size and complexity of the environment.

Traditional Computing Cons

3. **Limited Scalability:** Traditional computing environments are not easily scalable, meaning that they cannot easily be expanded to meet increased demand or new requirements without significant effort. For example, if additional capacity is needed in a traditional environment, it may require purchasing additional hardware or software licenses, which can be costly and time-consuming.
4. **Security Risks:** Traditional computing environments are vulnerable to security risks due to their lack of flexibility and scalability. These environments are typically designed for specific tasks and do not have the ability to adapt quickly to changing threats or demands from users or customers. Additionally, these environments are often difficult to monitor for suspicious activity due to their complexity and lack of visibility into user behavior.

Cloud Computing Pros

Cost Savings: It eliminates the need to purchase and maintain expensive hardware and software.

Instead, companies can use cloud-based services on a pay-as-you-go basis, allowing them to scale their usage up or down depending on their needs.

In addition, cloud benefit businesses to reduce their upfront costs and ensure that they are only paying for the resources they actually need.

Increased Productivity: Cloud computing can also increase productivity by allowing employees to access data and applications from any location with an internet connection.

This means that **employees can work from home or while travelling without having to worry about accessing the necessary files or programs.**

Additionally, cloud computing allows teams to collaborate in real time, which can improve communication and speed up the completion of tasks.

Cloud Computing Pros

Easier Software Updates

With cloud computing, **software updates are handled automatically by the provider**, so customers don't have to worry about manually installing new versions of their applications or operating systems.

So, this not only saves time but also ensures that customers always have access to the latest features and bug fixes available for their applications.

Also, It helps customers stay compliant with industry regulations as they don't have to worry about staying up to date with new versions of software that may be required for compliance purposes.

Cloud Computing Pros

Increased Reliability

Cloud computing is also more reliable than traditional IT infrastructure as cloud providers typically employ redundant servers and other measures to ensure that customer data is always accessible when needed.

Additionally, most cloud providers offer Service Level Agreements (SLAs) which guarantee a certain level of uptime for customer applications and data so that businesses can plan accordingly if there are any outages or disruptions in service due to maintenance or other issues beyond their control.

Cloud Computing Pros

Easy Scalability

Moreover, cloud computing offers seamless scalability, enabling businesses to effortlessly expand their resources such as storage space and processing power whenever necessary, without the need to invest in extra hardware or hire additional support staff.

This makes it easier for businesses to grow without having to worry about investing heavily in infrastructure upgrades each time they want to expand their operations or take on more customers/clients/projects etc.

Cloud Computing Cons

Security Issues

- One of the biggest drawbacks of cloud computing is the potential for security issues.
- Since data is stored on remote servers, it can be vulnerable to cyber-attacks or unauthorized access.
- Additionally, if a company's cloud service provider experiences a breach, all of their customers' data could be compromised.
- To mitigate these risks, companies must ensure that their cloud service provider has robust security measures in place.

Cloud Computing Cons

Interruptions in Service: There may be interruptions in service due to outages or other technical issues.

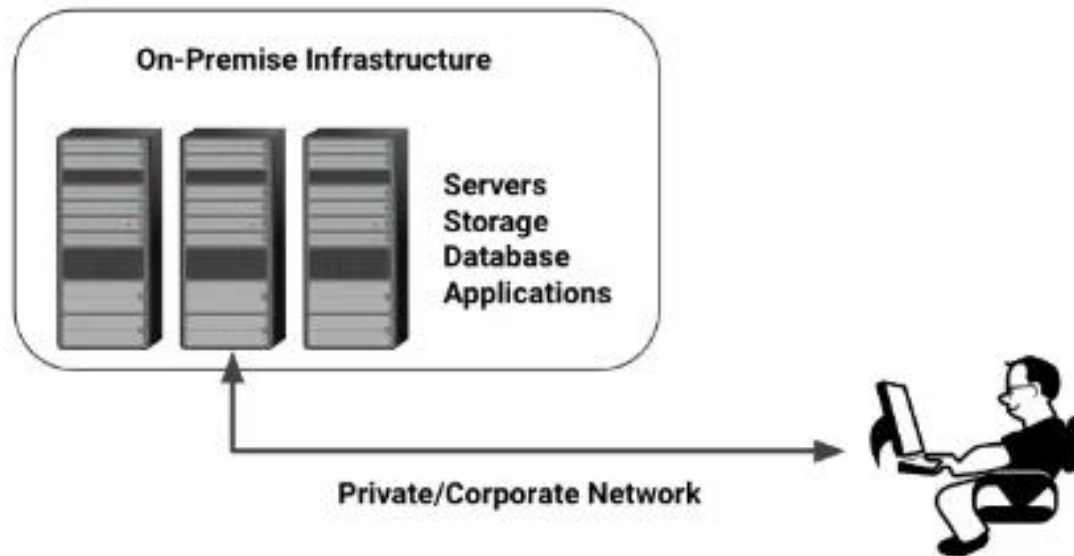
- If a company's cloud service provider experiences an outage or other issue, they may not be able to access their data until the problem is resolved.
- Additionally, if the internet connection between a company and its cloud service provider is unreliable, this could lead to frequent interruptions in service.

Lack of Control: Companies do not have full control over their data and applications since they are hosted by a third-party provider.

- This means that companies cannot customize their applications as much as they would like and may not always have access to the latest updates and features offered by the provider.

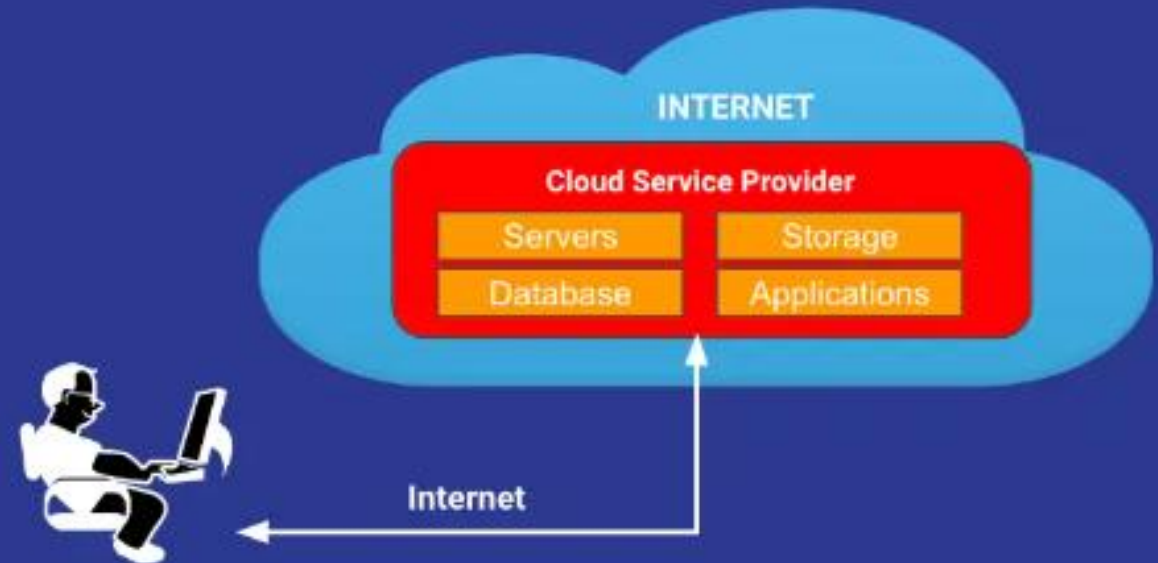
In One Glance

Traditional IT Infra



A traditional IT infrastructure is made up of the usual hardware and software components: **data centers, servers, storage, databases, networking hardware and enterprise applications**. IT Ops team will manage and update it when require.

Cloud Computing



A cloud computing service provides on-demand access to computer system resources, such as computing power, without requiring the user to actively manage them.



Model	Traditional Computing	Cloud Computing
Investment	› Invest in purchasing assets › Develop technical architecture › Add text here	› Invest in acquiring services › Architecture included › Add text here
Business	› Pay for cost of buying asset › Infrastructure cost in overhead › Add text here	› Pay for use › Reduce admin functions › Add text here
Delivery	› Costly, lengthy › Land and expand staffing › Add text here	› Faster return on investment › Reduce deployment cost › Add text here
Technical	› Non-shared › Static › Add text here	› Scalable and elastic › Dynamic › Add text here

Key Differences of Cloud Computing vs Traditional

Cost	Cloud computing can be much more cost-effective than traditional computing because it eliminates the need to purchase, install, and maintain hardware and software.
Scalability	• Cloud computing offers greater scalability than the traditional IT infrastructure. • With traditional computing, companies must purchase additional hardware and software in order to increase their capacity or add new features. • With cloud computing, however, companies can quickly scale up or down as needed without having to make any large investments in hardware or software.
Flexibility	• Cloud computing provides users with unparalleled flexibility as it enables them to effortlessly access their applications from any device, anywhere in the world, at any time, without the need for any prior installations on their devices. • This makes it easier for users to work remotely or collaborate with colleagues who may be located in different parts of the world without having to worry about compatibility issues or installation delays.

Key Differences of Cloud Computing vs Traditional

Security

- The security of cloud-based systems is also superior to traditional systems because cloud providers employ state-of-the-art security measures such as encryption and identity management tools that protect user data from unauthorized access or theft.
- Traditional systems are much more vulnerable to attack because they lack these advanced security measures and tend to rely on outdated methods of protection such as firewalls and antivirus software.

Model	Traditional Computing	Cloud Computing
Acquisition	• Buy Assets • Build Technical Architecture	• Buy Service • Architecture Included
Business	• Pay for Assets • Administrative Over Head	• Pay for use • Reduce admin function
Access	• Internal Networks • Corporate Desktops	• Over the internet • Any device
Technical	• Single Tenant, Non Shared • Static	• Multi-tenant, scalable, elastic • Dynamic
Delivery	• Costly, lengthy deployments • Land and Expand Staffing	• Reduce deployment tune • Fast ROI

Exercise Questions

1. Trace the history of cloud computing and explain how it evolved from earlier technologies like distributed computing.
2. Compare the computing trends of the past decades: mainframe computing, grid computing, and cloud computing. How has the focus shifted over time?
3. What advancements in technology have made cloud computing more feasible and popular in recent years?
4. Define fog computing and discuss its role in the Internet of Things (IoT).
5. How do the characteristics of cloud computing (on-demand self-service, broad network access, resource pooling, measured service) differentiate it from other computing paradigms?